

Again, computation of the eigenvalues λ_1 , λ_2 , and λ_3 of the symmetric coefficient matrix of the quadratic-form portion of Eq. (12) may give us quite a bit of information as to the type of quadric surface. However, actually executing the substitution and completing squares, which can be tedious, may be necessary to distinguish among certain surfaces. We give a rough classification scheme in Table 8.1. Remember that empty or degenerate cases are possible, even where we do not explicitly give them.

TABLE 8.1

Eigenvalues $\lambda_1, \lambda_2, \lambda_3$	Quadric Surface
All of the same sign	Ellipsoid
Two of one sign and one of the other sign	Elliptic cone, hyperboloid of two sheets, or hyperboloid of one sheet
One zero, two of the same sign	Elliptic paraboloid or elliptic cylinder (degenerate case)
One zero, two of opposite signs	Hyperbolic paraboloid or hyperbolic cylinder (degenerate case)
Two zero, one nonzero	Parabolic cylinder or two parallel planes (degenerate case)

If you try to verify Table 8.1, you will wonder about an equation of the form $ax^2 + by + cz = d$ in the last case given. Exercise 11 indicates that, by means of a rotation of axes, this equation can be reduced to $at_1^2 + rt_2 = d$, which can be written as $at_1^2 + r(t_2 - d/r) = 0$, and consequently describes a parabolic cylinder. We conclude with four examples.

EXAMPLE 3 Classify the quadric surface $2xy + 2xz = 1$.

SOLUTION Example 5 in Section 8.1 shows that the orthogonal substitution

$$\begin{aligned}x &= \frac{1}{\sqrt{2}}(t_2 - t_3) \\y &= \frac{1}{2}(-\sqrt{2}t_1 + t_2 + t_3) \\z &= \frac{1}{2}(\sqrt{2}t_1 + t_2 + t_3)\end{aligned}$$

transforms $2xy + 2xz$ into $\sqrt{2}t_2^2 - \sqrt{2}t_3^2$. It can be checked that the matrix C corresponding to this substitution has determinant 1. Thus, the substitution corresponds to a rotation of axes and transforms the equation $2xy + 2xz = 1$ into $\sqrt{2}t_2^2 - \sqrt{2}t_3^2 = 1$, which we recognize as a hyperbolic cylinder. $\square$

EXAMPLE 4 Classify the quadric surface $2xy + 2xz = y + 1$.

SOLUTION Using the same substitution as we did in Example 3, we obtain the equation

$$\sqrt{2}t_2^2 - \sqrt{2}t_3^2 = \frac{1}{2}(-\sqrt{2}t_1 + t_2 + t_3) + 1.$$

Translation of axes by completing squares yields an equation of the form

$$\sqrt{2}(t_2 - h)^2 - \sqrt{2}(t_3 - k)^2 = -\frac{1}{\sqrt{2}}(t_1 - r),$$

which we recognize as a hyperbolic paraboloid. ■

EXAMPLE 5 Classify the quadric surface $2xy + 2xz = x + 1$.

SOLUTION Using again the substitution in Example 3, we obtain

$$\sqrt{2}t_2^2 - \sqrt{2}t_3^2 = \frac{1}{\sqrt{2}}(t_2 - t_3) + 1$$

or

$$2t_2^2 - t_2 - 2t_3^2 + t_3 = \sqrt{2}.$$

Completing squares yields

$$2\left(t_2 - \frac{1}{4}\right)^2 - 2\left(t_3 - \frac{1}{4}\right)^2 = \sqrt{2},$$

which represents a hyperbolic cylinder. ■

EXAMPLE 6 Classify the quadric surface

$$2x^2 - 3y^2 + z^2 - 2xy + 4yz - 6x + 8y - 8z = 17$$

as far as possible, by finding just the eigenvalues of the symmetric coefficient matrix of the quadratic-form portion of the equation.

SOLUTION The symmetric matrix of the quadratic-form portion is

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -3 & 2 \\ 0 & 2 & 1 \end{bmatrix}.$$

We find that

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 2 - \lambda & -1 & 0 \\ -1 & -3 - \lambda & 2 \\ 0 & 2 & 1 - \lambda \end{vmatrix} \\ &= (2 - \lambda) \begin{vmatrix} -3 - \lambda & 2 \\ 2 & 1 - \lambda \end{vmatrix} + \begin{vmatrix} -1 & 2 \\ 0 & 1 - \lambda \end{vmatrix} \\ &= (2 - \lambda)(\lambda^2 + 2\lambda - 7) + \lambda - 1 \\ &= -\lambda^3 + 12\lambda - 15. \end{aligned}$$

We can see that $\lambda = 0$ is not a solution of the characteristic equation $-\lambda^3 + 12\lambda - 15 = 0$. We could plot a rough sketch of the graph of $y = -\lambda^3 + 12\lambda - 15$, just to determine the signs of the eigenvalues. However, we prefer to use the routine MATCOMP in LINTEK, or MATLAB, with the matrix A . We quickly find that the eigenvalues are approximately

$$\lambda_1 = -3.9720, \quad \lambda_2 = 1.5765, \quad \lambda_3 = 2.3954.$$

According to Table 8.1, we have an elliptic cone, a hyperboloid of two sheets, or a hyperboloid of one sheet. ■

SUMMARY

- Given an equation $ax^2 + bxy + cy^2 + dx + ey + f = 0$, let A be the symmetric coefficient matrix of the quadratic-form portion of the equation (the portion in color), let λ_1 and λ_2 be the eigenvalues of A , and let C be an orthogonal matrix of determinant 1 with eigenvectors of A for columns. The substitution corresponding to matrix C followed by translation of axes reduces the given equation to a standard form for the equation of a conic section. In particular, the equation describes a (possibly degenerate or empty)

ellipse if $\lambda_1\lambda_2 > 0$,

hyperbola if $\lambda_1\lambda_2 < 0$,

parabola if $\lambda_1\lambda_2 = 0$.

- Proceeding in a way analogous to that described in summary item (1) but for the equation

$$ax^2 + by^2 + cz^2 + dxy + exz + fyz + px + qy + rz + s = 0,$$

one obtains a standard form for the equation of a (possibly degenerate or empty) quadric surface in space. Table 8.1 lists the information that can be obtained from the three eigenvalues λ_1 , λ_2 , and λ_3 alone.

EXERCISES

In Exercises 1–8, rotate axes, using a substitution

$$\begin{bmatrix} x \\ y \end{bmatrix} = C \begin{bmatrix} t_1 \\ t_2 \end{bmatrix},$$

complete squares if necessary, and sketch the graph (if it is not empty) of the given conic section.

1. $2xy = 1$

2. $2xy - 2\sqrt{2}y = 1$

3. $x^2 + 2xy + y^2 = 4$

4. $x^2 - 2xy + y^2 + 4\sqrt{2}x = 4$

5. $10x^2 + 6xy + 2y^2 = 4$

6. $5x^2 + 4xy + 2y^2 = -1$

7. $3x^2 + 4xy + 6y^2 = 8$

8. $x^2 + 8xy + 7y^2 + 18\sqrt{5}x = -9$

9. Show that the plane curve $ax^2 + bxy + cy^2 + dx + ey + f = 0$ is a (possibly degenerate or empty)

ellipse if $b^2 - 4ac < 0$,

hyperbola if $b^2 - 4ac > 0$,

parabola if $b^2 - 4ac = 0$.

[HINT: Diagonalize $ax^2 + bxy + cy^2$, using the quadratic formula, and check the signs of the eigenvalues.]

10. Use Exercise 9 to classify the conic section with the given equation.

- $2x^2 + 8xy + 8y^2 - 3x + 2y = 13$
- $y^2 + 4xy - 5x^2 - 3x = 12$
- $-x^2 + 5xy - 7y^2 - 4y + 11 = 0$
- $xy + 4x - 3y = 8$
- $2x^2 - 3xy + y^2 - 8x + 5y = 30$
- $x^2 + 6xy + 9y^2 - 2x + 14y = 10$
- $4x^2 - 2xy - 3y^2 + 8x - 5y = 17$
- $8x^2 + 6xy + 2y^2 - 5x = 25$
- $x^2 - 2xy + 4x - 5y = 6$
- $2x^2 - 3xy + 2y^2 - 8y = 15$

11. Show that the equation $ax^2 + by + cz = d$ can be transformed into an equation of the form $at_1^2 + rt_2 = d$ by a rotation of axes in space. [HINT:

$$\text{If } \begin{bmatrix} x \\ y \\ z \end{bmatrix} = C \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}, \text{ then}$$

$$\begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} = C^{-1} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = C^T \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Find an orthogonal matrix C^T such that $\det(C^T) = 1$, $t_1 = x$, and $t_2 = (by + cz)/r$ for some r .]

In Exercises 12–20, classify the quadric surface with the given equation as one of the (possibly empty or degenerate) types illustrated in Figures 8.7–8.15.

12. $2x^2 + 2y^2 + 6yz + 10z^2 = 9$

13. $2xy + z^2 + 1 = 0$

14. $2xz + y^2 + 2y + 1 = 0$

15. $x^2 + 2yz + 4x + 1 = 0$

16. $3x^2 + 2y^2 + 6xz + 3z^2 = 1$

17. $x^2 - 8xy + 16y^2 - 3z^2 = 8$

18. $x^2 + 4y^2 - 4xz + 4z^2 = 8$

19. $-3x^2 + 2y^2 + 8yz + 16z^2 = 10$

20. $x^2 + y^2 + z^2 - 2xy + 2xz - 2yz = 9$

 In Exercises 21–27, use the routine *MATCOMP* in *LINTEK*, or *MATLAB*, to classify the quadric surface according to Table 8.1.

21. $x^2 + y^2 + z^2 - 2xy - 2xz - 2yz + 3x - 3z = 8$

22. $3x^2 + 2y^2 + 5z^2 + 4xy + 2yz - 3x + 10y = 4$

23. $2x^2 - 8y^2 + 3z^2 - 4xy + yz + 6xz - 3x - 8z = 3$

24. $3x^2 + 7y^2 + 4z^2 + 6xy - 8yz + 16x = 20$

25. $x^2 + 4y^2 + 16z^2 + 4xy + 8xz + 16yz - 8x + 3y = 8$

26. $x^2 + 6y^2 + 4z^2 - xy - 2xz - 3yz - 9x = 20$

27. $4x^2 - 3y^2 + z^2 + 8xz + 6yz + 2x - 3y = 8$

8.3

APPLICATIONS TO EXTREMA

Finding Extrema of Functions

We turn now to one of the applications of linear algebra to calculus. We simply state the facts from calculus that we need, trying to make them seem reasonable where possible.

There are many situations in which one desires to maximize or minimize a function of one or more variables. Applications involve maximizing profit, minimizing costs, maximizing speed, minimizing time, and so on. Such problems are of great practical importance.

Polynomial functions are especially easy to work with. Many important functions, such as trigonometric, exponential, and logarithmic functions, can't be expressed by polynomial formulas. However, each of these functions, and many others, can be expressed near a point in the domain of the function as a "polynomial" of infinite degree—an *infinite series*, as it is called. For example, it is shown in calculus that

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots \quad (1)$$

for any number x . [Recall that $2! = 2 \cdot 1$, $3! = 3 \cdot 2 \cdot 1$, and in general, $n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$.] We leave to calculus the discussion of such things as the interpretation of the infinite sum

$$1 - \frac{1}{2!} + \frac{1}{4!} - \frac{1}{6!} + \cdots$$

as $\cos 1$. From Eq. (1), it would seem that we should have

$$\cos(2x - y) = 1 - \frac{1}{2!}(2x - y)^2 + \frac{1}{4!}(2x - y)^4 - \frac{1}{6!}(2x - y)^6 + \cdots \quad (2)$$

Notice that the term

$$-\frac{1}{2!}(2x - y)^2 = -\frac{1}{2}(4x^2 - 4xy + y^2)$$

is a quadratic form—that is, a form (homogeneous polynomial) of degree 2. Similarly,

$$\frac{1}{4!}(2x - y)^4 = \frac{1}{24}(16x^4 - 32x^3y + 24x^2y^2 - 8xy^3 + y^4)$$

is a form of degree 4.

Consider a function $g(x_1, x_2, \dots, x_n)$ of n variables, which we denote as usual by $g(\mathbf{x})$. For many of the most common functions $g(\mathbf{x})$, it can be shown that, if $\mathbf{x}$ is near $\mathbf{0}$, so that all x_i are small in magnitude, then

$$g(\mathbf{x}) = c + f_1(\mathbf{x}) + f_2(\mathbf{x}) + f_3(\mathbf{x}) + \cdots + f_n(\mathbf{x}) + \cdots, \quad (3)$$

where each $f_i(\mathbf{x})$ is a form of degree i or is zero. Equation (2) illustrates this.

We wish to determine whether $g(\mathbf{x})$ in Eq. (3) has a *local extremum*—that is, a *local maximum* or *local minimum* at the origin $\mathbf{x} = \mathbf{0}$. A function $g(\mathbf{x})$ has a **local maximum at $\mathbf{0}$** if $g(\mathbf{x}) \leq g(\mathbf{0})$ for all $\mathbf{x}$ where $\|\mathbf{x}\|$ is sufficiently small. The notion of a **local minimum at $\mathbf{0}$** for $g(\mathbf{x})$ is analogously defined. For example, the function of one variable $g(x) = x^2 + 1$, whose graph is shown in Figure 8.16, has a local minimum of 1 at $x = 0$.

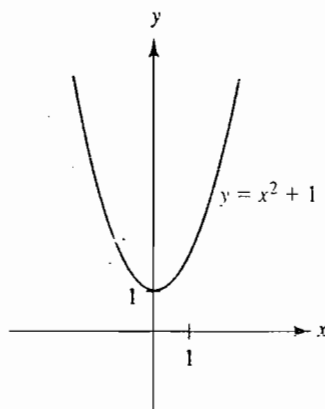


FIGURE 8.16

The graph of $y = g(x) = x^2 + 1$.

Now if x is really close to 0, all the forms $f_1(x)$, $f_2(x)$, $f_3(x)$, $\dots$ in Eq. (3) have very small values, so the constant c , if $c \neq 0$, is the dominant term of Eq. (3) near zero. Notice that, for $g(x) = 1 + x^2$ in Figure 8.16, the function has values close to the constant 1 for x close to zero.

After a nonzero constant, the form in Eq. (3) that contributes the most to $g(x)$ for $\|x\|$ close to 0 is the nonzero form $f_i(x)$ of lowest degree, because the lower the degree of a term of a polynomial, the more it contributes when the variables are all near zero. For example, x is greater than x^2 near zero; if $x = \frac{1}{100}$, then x^2 is only $\frac{1}{10,000}$. If $f_1(x) = 0$ and $f_2(x) \neq 0$, then $f_2(x)$ is the dominant form of Eq. (3) after a nonzero constant c , and so on.

We claim that, if $f_i(x) \neq 0$ in Eq. (3), then $g(x)$ does not have a local maximum or minimum at $x = 0$. Suppose that

$$f_i(x) = d_1x_1 + d_2x_2 + \dots + d_nx_n,$$

with some coefficient—say, d_1 —nonzero. If k is small but of the same sign as d_1 , then near the point $(k, 0, 0, \dots, 0)$ the function $g(x) \approx c + d_1k > c$. On the other hand, if k is small but of opposite sign to d_1 , then near this point $g(x) < c$. Thus, $g(x)$ can have a local maximum or minimum of c at 0 only if $f_i(x) = 0$ for all x .

If $f_1(x) = 0$ and $f_2(x) \neq 0$ in Eq. (3), then, for x near 0, $f_2(x)$ is the dominant form in the equation; it can be expected to dominate the terms of higher degree for x close to 0. It seems reasonable that, if $f_2(x) > 0$ for all $x \neq 0$ but near 0, then for such x

$$g(x) = c + (\text{Little bit}),$$

and $g(x)$ has a local minimum of c at 0. On the other hand, if $f_2(x) < 0$ for all such x , then we expect that

$$g(x) = c - (\text{Little bit})$$

for these values $\mathbf{x}$, and $g(\mathbf{x})$ has a local maximum of c at $\mathbf{0}$. This is proved in an advanced calculus course.

We know that we can orthogonally diagonalize the form $f_2(\mathbf{x})$ with a substitution $\mathbf{x} = C\mathbf{t}$ to become

$$\lambda_1 t_1^2 + \lambda_2 t_2^2 + \cdots + \lambda_n t_n^2, \quad (4)$$

where the λ_i are the eigenvalues of the symmetric coefficient matrix of $f_2(\mathbf{x})$. Form (4) is > 0 for all nonzero $\mathbf{t}$, and hence $f_2(\mathbf{x}) > 0$ for all nonzero $\mathbf{x}$, if and only if we have all $\lambda_i > 0$. Similarly, $f_2(\mathbf{x}) < 0$ for all nonzero $\mathbf{x}$ if and only if all $\lambda_i < 0$. It is also clear that, if some λ_i are positive and some are negative, then form (4) and hence $f_2(\mathbf{x})$ assume both positive and negative values arbitrarily close to zero.

DEFINITION 8.1 Definite Quadratic Forms

A quadratic form $f(\mathbf{x})$ is **positive definite** if $f(\mathbf{x}) > 0$ for all nonzero $\mathbf{x}$ in $\mathbb{R}^n$, and it is **negative definite** if $f(\mathbf{x}) < 0$ for all such nonzero $\mathbf{x}$.

Our work in Section 8.1 and the preceding statement give us the following theorem and corollary.

THEOREM 8.4 Criteria for Definite Quadratic Forms

A quadratic form is **positive definite** if and only if all the eigenvalues of its symmetric coefficient matrix are positive, and it is **negative definite** if and only if all those eigenvalues are negative.

COROLLARY A Test for Local Extrema

Let $g(\mathbf{x})$ be a function of n variables given by Eq. (3). Suppose that the function $f_1(\mathbf{x})$ in Eq. (3) is zero. If $f_2(\mathbf{x})$ is positive definite, then $g(\mathbf{x})$ has a local minimum of c at $\mathbf{x} = \mathbf{0}$, whereas if $f_2(\mathbf{x})$ is negative definite, then $g(\mathbf{x})$ has a local maximum of c at $\mathbf{x} = \mathbf{0}$. If $f_2(\mathbf{x})$ assumes both positive and negative values, then $g(\mathbf{x})$ has no local extremum at $\mathbf{x} = \mathbf{0}$.

We have discussed local extrema only at the origin. To do similar work at another point $(h_1, h_2, \dots, h_n)$, we need only translate axes to this point, letting $\bar{x}_i = x_i - h_i$ so that Eq. (3) becomes

$$g(\bar{\mathbf{x}}) = c + f_1(\bar{\mathbf{x}}) + f_2(\bar{\mathbf{x}}) + \cdots + f_n(\bar{\mathbf{x}}) + \cdots \quad (5)$$

Our discussion indicates a method for attempting to find local extrema of a function, which we box.

Finding an Extremum of $g(\mathbf{x})$

- Step 1** Find a point $\mathbf{h}$ where the function $f_i(\bar{\mathbf{x}}) = f_i(\mathbf{x} - \mathbf{h})$ in Eq. (5) becomes the zero function. (This is the province of calculus.)
- Step 2** Find the quadratic form $f_2(\bar{\mathbf{x}})$ at the point $\mathbf{h}$. (This also requires calculus.)
- Step 3** Find the eigenvalues of the symmetric coefficient matrix of the quadratic form.
- Step 4** If all eigenvalues are positive, then $g(\bar{\mathbf{x}})$ has a local minimum of c at $\mathbf{h}$. If all are negative, then $g(\bar{\mathbf{x}})$ has a local maximum of c at $\mathbf{h}$. If eigenvalues of both signs occur, then $g(\bar{\mathbf{x}})$ has no local extremum at $\mathbf{h}$.
- Step 5** If any of the eigenvalues is zero, further study is necessary.

Exercises 17 through 22 illustrate some of the things that can occur if step 5 is the case. We shall not tackle steps 1 or 2, because they require calculus, but will simply start with equations of the form of Eqs. (3) or (5) that meet the requirements in steps 1 and 2.

EXAMPLE 1 Let

$$g(x, y) = 3 + (2x^2 - 4xy + 4y^2) + (x^3 + 4xy^2 + y^3).$$

Determine whether $g(x, y)$ has a local extremum at the origin.

SOLUTION The symmetric coefficient matrix for the quadratic-form portion $2x^2 - 4xy + 4y^2$ of $g(x, y)$ is

$$A = \begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix}.$$

We find that

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & -2 \\ -2 & 4 - \lambda \end{vmatrix} = \lambda^2 - 6\lambda + 4.$$

The solutions of the characteristic equation $\lambda^2 - 6\lambda + 4 = 0$ are found by the quadratic formula to be

$$\lambda = \frac{6 \pm \sqrt{36 - 16}}{2} = 3 \pm \sqrt{5}.$$

We see that $\lambda_1 = 3 + \sqrt{5}$ and $\lambda_2 = 3 - \sqrt{5}$ are both positive, so our form is positive definite. Thus, $g(x, y)$ has a local minimum of 3 at $(0, 0)$. ■

EXAMPLE 2 Suppose that

$$g(x, y, z) = 7 + (2x^2 - 8y^2 + 3z^2 - 4xy + 2yz + 6xz) + (xz^2 - 5y^3) + (\text{higher-degree terms}).$$

Determine whether $g(x, y, z)$ has a local extremum at the origin.

SOLUTION The symmetric coefficient matrix of the quadratic-form portion is

$$A = \begin{bmatrix} 2 & -2 & 3 \\ -2 & -8 & 1 \\ 3 & 1 & 3 \end{bmatrix}.$$

We could find $p(\lambda) = |A - \lambda I|$ and attempt to solve the characteristic equation, or we could try to sketch the graph of $p(\lambda)$ well enough to determine the signs of the eigenvalues. However, we prefer to use the routine MATCOMP in LINTEK, or MATLAB, and we find that the eigenvalues are approximately

$$\lambda_1 = -8.605, \quad \lambda_2 = 0.042, \quad \lambda_3 = 5.563.$$

Because both positive and negative eigenvalues appear, there is no local extremum at the origin. ■

Maximizing or Minimizing a Quadratic Form on the Unit Sphere

Let $f(\mathbf{x})$ be a quadratic form in the variables $x_1, x_2, \dots, x_n$. We consider the problem of finding the maximum and minimum values of $f(\mathbf{x})$ for $\mathbf{x}$ on the *unit sphere*, where $\|\mathbf{x}\| = 1$ —that is, where $x_1^2 + x_2^2 + \dots + x_n^2 = 1$. It is shown in advanced calculus that such extrema of $f(\mathbf{x})$ on the unit sphere always exist.

We need only orthogonally diagonalize the form $f(\mathbf{x})$, using an orthogonal transformation $\mathbf{x} = C\mathbf{t}$, obtaining as usual

$$\lambda_1 t_1^2 + \lambda_2 t_2^2 + \dots + \lambda_n t_n^2. \quad (6)$$

Because our new basis for $\mathbb{R}^n$ is again orthonormal, the unit sphere has t -equation

$$t_1^2 + t_2^2 + \dots + t_n^2 = 1; \quad (7)$$

this is a very important point. Suppose that the λ_i are arranged so that

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n.$$

On the unit sphere with Eq. (7), we see that formula (6) can be written as

$$\begin{aligned} \lambda_1(1 - t_2^2 - \dots - t_n^2) + \lambda_2 t_2^2 + \dots + \lambda_n t_n^2 \\ = \lambda_1 - (\lambda_1 - \lambda_2)t_2^2 - \dots - (\lambda_1 - \lambda_n)t_n^2. \end{aligned} \quad (8)$$

Because $\lambda_1 - \lambda_i \geq 0$ for $i > 1$, we see that the maximum value assumed by formula (8) is λ_1 when $t_2 = t_3 = \dots = t_n = 0$, and $t_1 = \pm 1$. Exercise 32 indicates similarly that the minimum value assumed by form (6) is λ_n , when $t_n = \pm 1$ and all other $t_i = 0$. We state this as a theorem.

THEOREM 8.5 Extrema of a Quadratic Form on the Unit Sphere

Let $f(\mathbf{x})$ be a quadratic form, and let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of the symmetric coefficient matrix of $f(\mathbf{x})$. The maximum value assumed by $f(\mathbf{x})$ on the unit sphere $\|\mathbf{x}\| = 1$ is the maximum of the λ_i , and the minimum value assumed is the minimum of the λ_i . Each extremum is assumed at any eigenvector of length 1 corresponding to the eigenvalue that gives the extremum.

The preceding theorem is very important in vibration applications ranging from aerodynamics to particle physics. In such applications, one often needs to know the eigenvalue of maximum or of minimum magnitude for a symmetric matrix. These eigenvalues are frequently found by using advanced-calculus techniques to maximize or minimize the value of a quadratic form on a unit sphere, rather than using algebraic techniques such as those presented in this text for finding eigenvalues. The principal axis theorem (Theorem 8.1) and the preceding theorem are the algebraic foundation for such an analytic approach. We illustrate Theorem 8.5 with an example that maximizes a quadratic form on a unit sphere by finding the eigenvalues, rather than illustrating the more important reverse procedure.

EXAMPLE 3 Find the maximum and minimum values assumed by $2xy + 2xz$ on the unit sphere $x^2 + y^2 + z^2 = 1$, and find all points where these extrema are assumed.

SOLUTION From Example 5 in Section 8.1, we see that the eigenvalues of the symmetric coefficient matrix A and an associated eigenvector are given by

$$\begin{aligned}\lambda_1 &= 0, & \mathbf{v}_1 &= [0, -1, 1], \\ \lambda_2 &= \sqrt{2}, & \mathbf{v}_2 &= [\sqrt{2}, 1, 1], \\ \lambda_3 &= -\sqrt{2}, & \mathbf{v}_3 &= [-\sqrt{2}, 1, 1].\end{aligned}$$

We see that the maximum value assumed by $2xy + 2xz$ on the unit sphere is $\sqrt{2}$, and it is assumed at the points $\pm(\sqrt{2}/2, \frac{1}{2}, \frac{1}{2})$. The minimum value assumed is $-\sqrt{2}$, and it is assumed at the points $\pm(-\sqrt{2}/2, \frac{1}{2}, \frac{1}{2})$. Notice that we normalized our eigenvectors to length 1 so that they extend to the unit sphere. ■

The extension of Theorem 8.5 to a sphere centered at the origin, with radius other than 1, is left as Exercise 33.

SUMMARY

1. Let $g(\mathbf{x}) = c + f_1(\mathbf{x}) + f_2(\mathbf{x}) + \dots + f_n(\mathbf{x}) + \dots$ near $\mathbf{x} = 0$, where $f_i(\mathbf{x})$ is a form of degree i or is zero.
 - a. If $f_1(\mathbf{x}) = 0$ and $f_2(\mathbf{x})$ is positive definite, then $g(\mathbf{x})$ has a local minimum of c at $\mathbf{x} = 0$.

- b. If $f_1(\mathbf{x}) = 0$ and $f_2(\mathbf{x})$ is negative definite, then $g(\mathbf{x})$ has a local maximum of c at $\mathbf{x} = \mathbf{0}$.
- c. If $f_1(\mathbf{x}) \neq 0$ or if the symmetric coefficient matrix of $f_2(\mathbf{x})$ has both positive and negative eigenvalues, then $g(\mathbf{x})$ has no local extremum at $\mathbf{x} = \mathbf{0}$.
2. The natural analogue of summary item 1 holds at $\mathbf{x} = \mathbf{h}$; just translate the axes to the point $\mathbf{h}$, and replace x_i by $\bar{x}_i = x_i - h_i$.
3. A quadratic form in n variables has as maximum (minimum) value on the unit sphere $\|\mathbf{x}\| = 1$ in $\mathbb{R}^n$ the maximum (minimum) of the eigenvalues of the symmetric coefficient matrix of the form. The maximum (minimum) is assumed at each corresponding eigenvector of length 1.

EXERCISES

In Exercises 1–15, assume that $g(\mathbf{x})$, or $g(\bar{\mathbf{x}})$, is described by the given formula for values of $\mathbf{x}$, or $\bar{\mathbf{x}}$, near zero. Draw whatever conclusions are possible concerning local extrema of the function g .

- $g(x, y) = -7 + (3x^2 - 6xy + 4y^2) + (x^3 - 4y^3)$
- $g(x, y) = 8 - (2x^2 - 8xy + 3y^2) + (2x^2y - y^3)$
- $g(x, y) = 4 - 3x + (2x^2 - 2xy + y^2) + (2x^2y + y^3) + \dots$
- $g(x, y) = 5 - (8x^2 - 6xy + 2y^2) + (4x^3 - xy^2) + \dots$
- $g(\bar{x}, \bar{y}) = 3 - (4\bar{x}^2 - 8\bar{x}\bar{y} + 5\bar{y}^2) + (2\bar{x}^2\bar{y} - \bar{y}^3) + \dots$, $\bar{x} = x + 5$, $\bar{y} = y$
- $g(x, y) = -2 + (8x^2 + 4xy + y^2) + (x^3 + 5x^2y)$
- $g(x, y) = 5 + (3x^2 + 10xy + 7y^2) + (7xy^2 - y^3)$
- $g(x, y) = 4 - (x^2 - 6xy + 9y^2) + (x^3 - y^3) + \dots$
- $g(\bar{x}, \bar{y}) = 3 + (2\bar{x}^2 + 8\bar{x}\bar{y} + 8\bar{y}^2) + (4\bar{x}^3 - \bar{x}\bar{y}^2) + \dots$, $\bar{x} = x - 3$, $\bar{y} = y - 1$
- $g(\bar{x}, \bar{y}) = 4 - (\bar{x}^2 + 3\bar{x}\bar{y} - \bar{y}^2) + (4\bar{x}^2\bar{y} - 5\bar{x}\bar{y}^2)$, $\bar{x} = x + 1$, $\bar{y} = y - 7$
- $g(x, y, z) = 4 - (x^2 + 4xy + 5y^2 + 3z^2) + (x^3 - xyz)$
- $g(x, y, z) = 3 + (2x^2 + 6xz - y^2 + 5z^2) + (x^2z - y^2z) + \dots$
- $g(x, y, z) = 5 + (4x^2 + 2xy + y^2 - z^2) + (xy^2 - 4xyz) + \dots$
- $g(\bar{x}, \bar{y}, \bar{z}) = 4 + (\bar{x}^2 + \bar{y}^2 + \bar{z}^2 - 2\bar{x}\bar{z} - 2\bar{x}\bar{y} - 2\bar{y}\bar{z}) + (3\bar{x}^3 - \bar{z}^3) + \dots$, $\bar{x} = x + 1$, $\bar{y} = y - 2$, $\bar{z} = z + 5$
- $g(\bar{x}, \bar{y}, \bar{z}) = 4 - 3\bar{z} + (2\bar{x}^2 - 2\bar{x}\bar{y} + 3\bar{x}\bar{z} + 5\bar{y}^2 + \bar{z}^2) + (2\bar{x}\bar{y}\bar{z} - \bar{z}^3) + \dots$, $\bar{x} = x - 7$, $\bar{y} = y + 6$, $\bar{z} = z$
- Define the notion of a local minimum c of a function $f(\mathbf{x})$ of n variables at a point $\mathbf{h}$.

In Exercises 17–22, let

$$g(x, y) = c + f_1(x, y) + f_2(x, y) + f_3(x, y) + \dots$$

in accordance with the notation we have been using. Let λ_1 and λ_2 be the eigenvalues of the symmetric coefficient matrix of $f_2(x, y)$.

- Give an example of a polynomial function $g(x, y)$ for which $f_1(x, y) = 0$, $\lambda_1 = 0$, $|\lambda_2| = 1$, and $g(x, y)$ has a local minimum of 10 at the origin.
- Repeat Exercise 17, but make $g(x, y)$ have a local maximum of -5 at the origin.
- Repeat Exercise 17, but make $g(x, y)$ have no local extremum at the origin.
- Give an example of a polynomial function $g(x, y)$ such that $f_1(x, y) = f_2(x, y) = 0$, having a local maximum of 1 at the origin.

21. Repeat Exercise 20, but make $g(x, y)$ have a local minimum of 40 at the origin.
22. Repeat Exercise 20, but make $g(x, y)$ have no local extremum at the origin.

In Exercises 23–31, find the maximum and minimum values of the quadratic form on the unit circle in $\mathbb{R}^2$, or unit sphere in $\mathbb{R}^n$, and find all points on the unit circle or sphere where the extrema are assumed.

23. $xy, n = 2$
24. $3x^2 + 4xy, n = 2$
25. $-6xy + 8y^2, n = 2$
26. $x^2 + 2xy + y^2, n = 3$
27. $3x^2 - 6xy + 3y^2, n = 2$
28. $y^2 + 2xz, n = 3$
29. $x^2 + y^2 + z^2 - 2xy + 2xz - 2yz, n = 3$
30. $x^2 + y^2 + z^2 - 2xy - 2xz - 2yz, n = 3$
[SUGGESTION: Use Example 4 in Section 6.3.]
31. $x^2 + w^2 + 4yz - 2xw, n = 4$
32. Prove that the minimum value assumed by a quadratic form in n variables on the unit sphere in $\mathbb{R}^n$ is the minimum eigenvalue of the symmetric coefficient matrix of the form,

and prove that this minimum value is assumed at any corresponding eigenvector of length 1.

33. Let $f(\mathbf{x})$ be a quadratic form in n variables. Describe the maximum and minimum values assumed by $f(\mathbf{x})$ on the sphere $\|\mathbf{x}\| = a^2$ for any $a > 0$. Describe the points of the sphere at which these extrema are assumed. [HINT: For any k in $\mathbb{R}$, how does $f(k\mathbf{x})$ compare with $f(\mathbf{x})$?



In Exercises 34–38, use the routine MATCOMP in LINTEK, or MATLAB, and follow the directions for Exercises 1–15.

34. $g(x, y, z) = 7 + (3x^2 + 2y^2 + 5z^2 + 4xy + 2yz) + (xyz - x^2y)$
35. $g(x, y, z) = 5 - (x^2 + 6y^2 + 4z^2 - xy - 2xz - 3yz) + (xz^2 - 5z^3) + \dots$
36. $g(x, y, z) = -1 - (2x^2 - 8y^2 + 3z^2 - 4xy + yz + 6xz) + (8x^3 - 4y^2z + 3z^3)$
37. $g(x, y, z) = 7 + (x^2 + 2y^2 + z^2 - 3xy - 4xz - 3yz) + (x^2y - 4yz^2)$
38. $g(x, y, z) = 5 + (x^2 + 4y^2 + 16z^2 + 4xy + 8xz + 16yz) + (x^3 + y^3 + 7xyz) + \dots$

8.4

COMPUTING EIGENVALUES AND EIGENVECTORS

Computing the eigenvalues of a matrix is one of the toughest jobs in linear algebra. Many algorithms have been developed, but no one method can be considered the best for all cases. We have used the characteristic equation for computation of eigenvalues in most of our examples and exercises. The routine MATCOMP in LINTEK also uses it. Professionals frown on the use of this method in practical applications because small errors made in computing coefficients of the characteristic polynomial can lead to significant errors in eigenvalues. We describe three other methods in this section.

1. The *power method* is especially useful if one wants only the eigenvalue of largest (or of smallest) magnitude, as in many vibration problems.
2. *Jacobi's method* for symmetric matrices is presented without proof. This method is chiefly of historical interest, because the third and most recent of the methods we describe is more general and usually more efficient.

3. The *QR method* was developed by H. Rutishauser in 1958 and J. G. F. Francis in 1961 and is probably the most widely used method today. It finds all eigenvalues, both real and complex, of a real matrix. Details are beyond the scope of this text. We give only a rough idea of the method, with no proofs.

The routines POWER, JACOBI, and QRFACTOR in LINTEK can be used to illustrate the three methods.

The Power Method

Let A be an $n \times n$ diagonalizable matrix with real eigenvalues. Suppose that one eigenvalue—say, λ_1 —has greater magnitude than all the others. That is, $|\lambda_1| > |\lambda_i|$ for $i > 1$. We call λ_1 the **dominant eigenvalue** of A . In many vibration problems, we are interested only in computing this dominant eigenvalue and the eigenvalue of minimum absolute value.

Because A is diagonalizable, there exists a basis

$$\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$$

for $\mathbb{R}^n$ composed of eigenvectors of A . We assume that $\mathbf{b}_i$ is the eigenvector corresponding to λ_i , and that the numbering is such that $|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_n|$.

Let $\mathbf{w}_1$ be any nonzero vector in $\mathbb{R}^n$. Then

$$\mathbf{w}_1 = c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2 + \dots + c_n \mathbf{b}_n \quad (1)$$

for some constants c_i in $\mathbb{R}$. Applying A^s to both sides of Eq. (1) and remembering that $A^s \mathbf{b}_i = \lambda_i^s \mathbf{b}_i$, we see that

$$A^s \mathbf{w}_1 = \lambda_1^s c_1 \mathbf{b}_1 + \lambda_2^s c_2 \mathbf{b}_2 + \dots + \lambda_n^s c_n \mathbf{b}_n. \quad (2)$$

Because λ_1 is dominant, we see that, for large s , the summand $\lambda_1^s c_1 \mathbf{b}_1$ dominates the right-hand side of Eq. (2), as long as $c_1 \neq 0$. This is even more evident if we rewrite Eq. (2) in the form

$$A^s \mathbf{w}_1 = \lambda_1^s (c_1 \mathbf{b}_1 + (\lambda_2/\lambda_1)^s c_2 \mathbf{b}_2 + \dots + (\lambda_n/\lambda_1)^s c_n \mathbf{b}_n). \quad (3)$$

If s is large, the quotients $(\lambda_i/\lambda_1)^s$ for $i > 1$ are close to zero, because $|\lambda_i/\lambda_1| < 1$. Thus, if $c_1 \neq 0$ and s is large enough, $A^s \mathbf{w}_1$ is very nearly parallel to $\lambda_1^s c_1 \mathbf{b}_1$, which is an eigenvector of A corresponding to the eigenvalue λ_1 . This suggests that we can approximate an eigenvector of A corresponding to the dominant eigenvalue λ_1 by multiplying an appropriate initial approximation vector $\mathbf{w}_1$ repeatedly by A .

A few comments are in order. In a practical application, we may have a rough idea of an eigenvector for λ_1 and be able to choose a reasonably good first approximation $\mathbf{w}_1$. In any case, $\mathbf{w}_1$ *should not* be in the subspace of $\mathbb{R}^n$

generated by the eigenvectors corresponding to the λ_j for $j > 1$. This corresponds to the requirement that $c_1 \neq 0$.

Repeated multiplication of $\mathbf{w}_1$ by A may produce very large (or very small) numbers. It is customary to scale after each multiplication, to keep the components of the vectors at a reasonable size. After the first multiplication, we find the maximum d_1 of the magnitudes of all the components of $A\mathbf{w}_1$ and apply A the next time to the vector $\mathbf{w}_2 = (1/d_1)A\mathbf{w}_1$. Similarly, we let $\mathbf{w}_3 = (1/d_2)A\mathbf{w}_2$, where d_2 is the maximum of the magnitudes of components of $A\mathbf{w}_2$, and so on. Thus we are always multiplying A times a vector $\mathbf{w}_j$ with components of maximum magnitude 1. This scaling also aids us in estimating the number-of-significant-figures accuracy we have attained in the components of our approximations to an eigenvector.

If $\mathbf{x}$ is an eigenvector corresponding to λ_1 , then

$$\frac{A\mathbf{x} \cdot \mathbf{x}}{\mathbf{x} \cdot \mathbf{x}} = \frac{\lambda_1 \mathbf{x} \cdot \mathbf{x}}{\mathbf{x} \cdot \mathbf{x}} = \lambda_1. \quad (4)$$

The quotient $(A\mathbf{x} \cdot \mathbf{x})/(\mathbf{x} \cdot \mathbf{x})$ is called a **Rayleigh quotient**. As we compute the $\mathbf{w}_j$, the Rayleigh quotients $(A\mathbf{w}_j \cdot \mathbf{w}_j)/(\mathbf{w}_j \cdot \mathbf{w}_j)$ should approach λ_1 .

This *power method* for finding the dominant eigenvector should, mathematically, break down if we choose the initial approximation $\mathbf{w}_1$ in Eq. (1) in such a way that the coefficient c_1 of $\mathbf{b}_1$ is zero. However, due to roundoff error, it often happens that a nonzero component of $\mathbf{b}_1$ creeps into the $\mathbf{w}_j$ as they are computed, and the $\mathbf{w}_j$ then start swinging toward an eigenvector for λ_1 as desired. This is one case where roundoff error is helpful!

Equation (3) indicates that the ratio $|\lambda_2/\lambda_1|$, which is the maximum of the magnitudes $|\lambda_i/\lambda_1|$ for $i > 1$, should control the speed of convergence of the $\mathbf{w}_j$ to an eigenvector. If $|\lambda_2/\lambda_1|$ is close to 1, convergence may be quite slow.

We summarize the steps of the power method in the following box.

HISTORICAL NOTE THE RAYLEIGH QUOTIENT is named for John William Strutt, the third Baron Rayleigh (1842–1919). Rayleigh was a hereditary peer who surprised his family by pursuing a scientific career instead of contenting himself with the life of a country gentleman. He set up a laboratory at the family seat in Terling Place, Essex, and spent most of his life there pursuing his research into many aspects of physics—in particular, sound and optics. He is especially famous for his resolution of the long-standing question in optics as to why the sky is blue, as well as for his codiscovery of the element argon, for which he won the Nobel prize in 1904. When he received the British Order of Merit in 1902 he said that “the only merit of which he personally was conscious was that of having pleased himself by his studies, and any results that may have been due to his researches were owing to the fact that it had been a pleasure to him to become a physicist.”

Rayleigh used the Rayleigh quotient early in his career in an 1873 work in which he needed to evaluate approximately the normal modes of a complex vibrating system. He subsequently used it and related methods in his classic text *The Theory of Sound* (1877).

The Power Method for Finding the Dominant Eigenvalue λ_1 of A

- Step 1** Choose an appropriate vector $\mathbf{w}_1$ in $\mathbb{R}^n$ as first approximation to an eigenvector corresponding to λ_1 .
- Step 2** Compute $A\mathbf{w}_1$ and the Rayleigh quotient $(A\mathbf{w}_1 \cdot \mathbf{w}_1)/(\mathbf{w}_1 \cdot \mathbf{w}_1)$.
- Step 3** Let $\mathbf{w}_2 = (1/d_1)A\mathbf{w}_1$, where d_1 is the maximum of the magnitudes of components of $A\mathbf{w}_1$.
- Step 4** Repeat step 2, with all subscripts increased by 1. The Rayleigh quotients should approach λ_1 , and the $\mathbf{w}_j$ should approach an eigenvector of A corresponding to λ_1 .

EXAMPLE 1 Illustrate the power method for the matrix

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 0 \end{bmatrix} \text{ starting with } \mathbf{w}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

finding $\mathbf{w}_2$, $\mathbf{w}_3$, and the first two Rayleigh quotients.

SOLUTION We have

$$A\mathbf{w}_1 = \begin{bmatrix} 3 & -2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}.$$

We find as first Rayleigh quotient

$$\frac{A\mathbf{w}_1 \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1} = \frac{7}{2} = 3.5.$$

Because 5 is the maximum magnitude of a component of $A\mathbf{w}_1$, we have

$$\mathbf{w}_2 = \frac{1}{5}A\mathbf{w}_1 = \begin{bmatrix} 1 \\ -2/5 \end{bmatrix}.$$

Then

$$A\mathbf{w}_2 = \begin{bmatrix} 3 & -2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2/5 \end{bmatrix} = \begin{bmatrix} 19/5 \\ -2 \end{bmatrix}.$$

The next Rayleigh quotient is

$$\frac{A\mathbf{w}_2 \cdot \mathbf{w}_2}{\mathbf{w}_2 \cdot \mathbf{w}_2} = \frac{\frac{23}{5}}{\frac{29}{25}} = \frac{115}{29} \approx 3.966.$$

Finally,

$$\mathbf{w}_3 = \frac{5}{19} \begin{bmatrix} 19/5 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -10/19 \end{bmatrix}.$$

EXAMPLE 2 Use the routine **POWER** in **LINTEK** with the data in Example 1, and give the vectors w_j and the Rayleigh quotients until stabilization to all decimal places printed occurs.

SOLUTION **POWER** prints fewer significant figures for the vectors to save space. The data obtained using **POWER** are shown in Table 8.2. It is easy to solve the characteristic equation $\lambda^2 - 3\lambda - 4 = 0$ of A , and to see that the eigenvalues are really $\lambda_1 = 4$ and $\lambda_2 = -1$. **POWER** found the dominant eigenvalue 4, and it shows that $[1, -0.5]$ is an eigenvector for this eigenvalue. ■

If A has an eigenvalue of nonzero magnitude *smaller* than that of any other eigenvalue, the power method can be used with A^{-1} to find this smallest eigenvalue. The eigenvalues of A^{-1} are the reciprocals of the eigenvalues of A , and the eigenvectors are the same. To illustrate, if 5 is the dominant eigenvalue of A^{-1} and v is an associated eigenvector, then $\frac{1}{5}$ is the eigenvalue of A of smallest magnitude, and v is still an associated eigenvector.

TABLE 8.2

Power Method for $A = \begin{bmatrix} 3 & -2 \\ -2 & 0 \end{bmatrix}$

Vector Approximations	Rayleigh Quotients
$[1, -1]$	
$[1, -.4]$	3.5
$[1, -.5263158]$	3.96551724137931
$[1, -.4935065]$	3.997830802603037
$[1, -.5016286]$	3.999864369998644
$[1, -.4995932]$	3.999991522909338
$[1, -.5001018]$	3.999999470180991
$[1, -.4999746]$	3.999999966886309
$[1, -.5000064]$	3.99999997930394
$[1, -.4999984]$	3.9999999987065
$[1, -.5000004]$	3.99999999991916
$[1, -.4999999]$	3.99999999999495
$[1, -.5]$	3.99999999999968
$[1, -.5]$	3.99999999999998
$[1, -.5]$	4
$[1, -.5]$	4

Deflation for Symmetric Matrices

The method of deflation gives a way to compute eigenvalues of intermediate magnitude of a *symmetric* matrix by the power method. It is based on an

interesting decomposition of a matrix product AB . Let A be an $m \times n$ matrix, and let B be an $n \times s$ matrix. We write AB symbolically as

$$AB = \begin{bmatrix} | & | & & | \\ \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \text{---} \mathbf{r}_1 \text{---} \\ \text{---} \mathbf{r}_2 \text{---} \\ \vdots \\ \text{---} \mathbf{r}_n \text{---} \end{bmatrix},$$

where $\mathbf{c}_j$ is the j th column vector of A and $\mathbf{r}_i$ is the i th row vector of B . We claim that

$$AB = \mathbf{c}_1 \mathbf{r}_1 + \mathbf{c}_2 \mathbf{r}_2 + \cdots + \mathbf{c}_n \mathbf{r}_n, \quad (5)$$

where each $\mathbf{c}_i \mathbf{r}_i$ is the product of an $m \times 1$ matrix with a $1 \times s$ matrix. To see this, remember that the entry in the i th row and j th column of AB is

$$a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

But $\mathbf{c}_i \mathbf{r}_i$ contributes precisely $a_{i1}b_{1j}$ to the i th row and j th column of the sum in Eq. (5), while $\mathbf{c}_2 \mathbf{r}_2$ contributes $a_{i2}b_{2j}$, and so on. This establishes Eq. (5).

Let A be an $n \times n$ symmetric matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, where

$$|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n|.$$

We know from Section 6.3 that there exists an orthogonal matrix C such that $C^{-1}AC = D$, where D is a diagonal matrix, with $d_{ii} = \lambda_i$. Then

$$A = CDC^{-1} = CDC^T$$

$$= \begin{bmatrix} | & | & & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & 0 \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix} \begin{bmatrix} \text{---} \mathbf{b}_1^T \text{---} \\ \text{---} \mathbf{b}_2^T \text{---} \\ \vdots \\ \text{---} \mathbf{b}_n^T \text{---} \end{bmatrix}, \quad (6)$$

where $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ is an orthonormal basis for $\mathbb{R}^n$ consisting of eigenvectors of A and forming the column vectors of C . From Eq. (6), we see that

$$A = \begin{bmatrix} | & | & & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \text{---} \lambda_1 \mathbf{b}_1^T \text{---} \\ \text{---} \lambda_2 \mathbf{b}_2^T \text{---} \\ \vdots \\ \text{---} \lambda_n \mathbf{b}_n^T \text{---} \end{bmatrix}. \quad (7)$$

Using Eq. (5) and writing the $\mathbf{b}_i$ as column vectors, we see that A can be written in the following form:

Spectral Decomposition of A

$$A = \lambda_1 \mathbf{b}_1 \mathbf{b}_1^T + \lambda_2 \mathbf{b}_2 \mathbf{b}_2^T + \cdots + \lambda_n \mathbf{b}_n \mathbf{b}_n^T. \quad (8)$$

Equation (8) is known as the **spectral theorem** for symmetric matrices.

EXAMPLE 3 Illustrate the spectral theorem for the matrix

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 0 \end{bmatrix}$$

of Example 1.

SOLUTION Computation shows that the matrix A has eigenvalues $\lambda_1 = 4$ and $\lambda_2 = -1$, with corresponding eigenvectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}.$$

Eigenvectors $\mathbf{b}_1$ and $\mathbf{b}_2$ forming an orthonormal basis are

$$\mathbf{b}_1 = \begin{bmatrix} 2/\sqrt{5} \\ -1/\sqrt{5} \end{bmatrix} \quad \text{and} \quad \mathbf{b}_2 = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}.$$

We have

$$\begin{aligned} \lambda_1 \mathbf{b}_1 \mathbf{b}_1^T + \lambda_2 \mathbf{b}_2 \mathbf{b}_2^T &= 4 \begin{bmatrix} 2/\sqrt{5} \\ -1/\sqrt{5} \end{bmatrix} \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \end{bmatrix} - \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \\ &= 4 \begin{bmatrix} \frac{4}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{bmatrix} - \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{4}{5} \end{bmatrix} \\ &= \begin{bmatrix} \frac{16}{5} & -\frac{8}{5} \\ -\frac{8}{5} & \frac{4}{5} \end{bmatrix} - \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{4}{5} \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -2 & 0 \end{bmatrix} = A. \end{aligned}$$

HISTORICAL NOTE THE TERM *SPECTRUM* was coined around 1905 by David Hilbert (1862–1943) for use in dealing with the eigenvalues of quadratic forms in infinitely many variables. The notion of such forms came out of his study of certain linear operators in spaces of functions. Hilbert was struck by the analogy one could make between these operators and quadratic forms in finitely many variables. Hilbert's approach was greatly expanded and generalized during the next decade by Erhard Schmidt, Frigyes Riesz (1880–1956), and Hermann Weyl. Interestingly enough, in the 1920s physicists called on spectra of certain linear operators to explain optical spectra.

David Hilbert was the most influential mathematician of the early twentieth century. He made major contributions in many fields, including algebraic forms, algebraic number theory, integral equations, the foundations of geometry, theoretical physics, and the foundations of mathematics. His speech at the 2nd International Congress of Mathematicians in Paris in 1900, outlining the important mathematical problems of the day, proved extremely significant in providing the direction for twentieth-century mathematics.

Suppose now that we have found the eigenvalue λ_1 of maximum magnitude of A , and a corresponding eigenvector $\mathbf{v}_1$ by the power method. We compute the unit vector $\mathbf{b}_1 = \mathbf{v}_1/\|\mathbf{v}_1\|$. From Eq. (8), we see that

$$A - \lambda_1 \mathbf{b}_1 \mathbf{b}_1^T = 0 \mathbf{b}_1 \mathbf{b}_1^T + \lambda_2 \mathbf{b}_2 \mathbf{b}_2^T + \cdots + \lambda_n \mathbf{b}_n \mathbf{b}_n^T \quad (9)$$

is a matrix with eigenvalues $\lambda_2, \lambda_3, \dots, \lambda_n, 0$ in order of descending magnitude, and with corresponding eigenvectors $\mathbf{b}_2, \mathbf{b}_3, \dots, \mathbf{b}_n, \mathbf{b}_1$. We can now use the power method on this matrix to compute λ_2 and $\mathbf{b}_2$. We then execute this *deflation* again, forming $A - \lambda_1 \mathbf{b}_1 \mathbf{b}_1^T - \lambda_2 \mathbf{b}_2 \mathbf{b}_2^T$ to find λ_3 and $\mathbf{b}_3$, and so on.

The routine POWER in LINTEK has an option to use this method of deflation. For the symmetric matrices of the small size that we use in our examples and exercises, POWER handles deflation well, provided that we compute each eigenvalue until stabilization to all places shown on the screen is achieved. In practice, scientists are wary of using deflation to find more than one or two further eigenvalues, because any error made in computation of an eigenvalue or eigenvector will propagate errors in the computation of subsequent ones.

EXAMPLE 4 Illustrate Eq. (9) for deflation with the matrix

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 0 \end{bmatrix}$$

of Example 3.

SOLUTION From Example 3, we have

$$A = \lambda_1 \mathbf{b}_1 \mathbf{b}_1^T + \lambda_2 \mathbf{b}_2 \mathbf{b}_2^T = 4 \begin{bmatrix} \frac{4}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{bmatrix} - \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{4}{5} \end{bmatrix}.$$

Thus,

$$A - \lambda_1 \mathbf{b}_1 \mathbf{b}_1^T = - \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{4}{5} \end{bmatrix} = \begin{bmatrix} -\frac{1}{5} & -\frac{2}{5} \\ -\frac{2}{5} & -\frac{4}{5} \end{bmatrix}.$$

The characteristic polynomial for this matrix is

$$\begin{vmatrix} -\frac{1}{5} - \lambda & -\frac{2}{5} \\ -\frac{2}{5} & -\frac{4}{5} - \lambda \end{vmatrix} = (\lambda^2 + \lambda) = \lambda(\lambda + 1),$$

and the eigenvalues are indeed $\lambda_2 = -1$ together with 0, as indicated following Eq. (9). ■

EXAMPLE 5 Use the routine **POWER** in **LINTEK** to find the eigenvalues and eigenvectors, using deflation, for the symmetric matrix

$$\begin{bmatrix} 2 & -8 & 5 \\ -8 & 0 & 10 \\ 5 & 10 & -6 \end{bmatrix}.$$

SOLUTION Using **POWER** with deflation and finding eigenvalues as accurately as the printing on the screen permits, we obtain the eigenvectors and eigenvalues

$$\begin{aligned} \mathbf{v}_1 &= [-.6042427, -.8477272, 1], & \lambda_1 &= -17.49848531152027, \\ \mathbf{v}_2 &= [-.760632, 1, .3881208], & \lambda_2 &= 9.96626448890372, \\ \mathbf{v}_3 &= [1, .3958651, .9398283], & \lambda_3 &= 3.532220822616553. \end{aligned}$$

Using **MATCOMP** as a check, we find the same eigenvalues and eigenvectors. ■

Jacobi's Method for Symmetric Matrices

We present Jacobi's method for diagonalizing a symmetric matrix, omitting proofs. Let $A = [a_{ij}]$ be an $n \times n$ symmetric matrix, and suppose that a_{pq} is an entry of maximum magnitude among the entries of A that lie above the main diagonal. For example, in the matrix

$$A = \begin{bmatrix} 2 & -8 & 5 \\ -8 & 0 & 10 \\ 5 & 10 & -6 \end{bmatrix}, \quad (10)$$

the entry above the diagonal having maximum magnitude is 10. Then form the 2×2 matrix

$$\begin{bmatrix} a_{pp} & a_{pq} \\ a_{qp} & a_{qq} \end{bmatrix}. \quad (11)$$

From matrix (10), we would form the matrix consisting of the portion shown in color—namely,

$$\begin{bmatrix} 0 & 10 \\ 10 & -6 \end{bmatrix}.$$

Let $C = [c_{ij}]$ be a 2×2 orthogonal matrix that diagonalizes matrix (11). (Recall that C can always be chosen to correspond to a rotation of the plane, although this is not essential in order for Jacobi's method to work.) Now form an $n \times n$ matrix R , the same size as the matrix A , which looks like the identity matrix except that $r_{pp} = c_{11}$, $r_{pq} = c_{12}$, $r_{qp} = c_{21}$, and $r_{qq} = c_{22}$. For matrix (10), where 10 has maximum magnitude above the diagonal, we would have

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_{11} & c_{12} \\ 0 & c_{21} & c_{22} \end{bmatrix}.$$

This matrix R will be an orthogonal matrix, with $\det(R) = \det(C)$. Now form the new symmetric matrix $B_1 = R^T A R$, which has zero entries in the row p , column q position and in the row q , column p position. Other entries in B_1 can also be changed from those in A , but it can be shown that the maximum magnitude of off-diagonal entries has been reduced, assuming that no other above-diagonal entry in A had the magnitude of a_{pq} . Then repeat this process, starting with B_1 instead of A , to obtain another symmetric matrix B_2 , and so on. It can be shown that the maximum magnitude of off-diagonal entries in the matrices B_i approaches zero as i increases. Thus the sequence of matrices

$$B_1, B_2, B_3, \dots$$

will approach a diagonal matrix D whose eigenvalues $d_{11}, d_{22}, \dots, d_{nn}$ are the same as those of A .

If one is going to use Jacobi's method much, one should find the 2×2 matrix C that diagonalizes a general 2×2 symmetric matrix

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix};$$

that is, one should find formulas for computing C in terms of the entries a , b , and c . Exercise 23 develops such formulas.

Rather than give a tedious pencil-and-paper example of Jacobi's method, we choose to present data generated by the routine JACOBI in LINTEK for matrix (10)—which is the same matrix as that in Example 5, where we used the power method. Observe how, in each step, the colored entries of maximum magnitude off the diagonal are reduced to "zero." Although they may not remain zero in the next step, they never return to their original size.

EXAMPLE 6 Use the routine JACOBI in LINTEK to diagonalize the matrix

$$\begin{bmatrix} 2 & -8 & 5 \\ -8 & 0 & 10 \\ 5 & 10 & -6 \end{bmatrix}.$$

SOLUTION The routine JACOBI gives the following matrices:

$$\begin{bmatrix} 2 & -8 & 5 \\ -8 & 0 & 10 \\ 5 & 10 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 8.786909 & 3.433691 \\ 8.786909 & -13.44031 & 0 \\ 3.433691 & 0 & 7.440307 \end{bmatrix}$$

$$\begin{bmatrix} -17.41676 & 0 & -1.415677 \\ 0 & 5.97645 & -3.128273 \\ -1.415677 & -3.128273 & 7.440307 \end{bmatrix}$$

$$\begin{bmatrix} -17.41676 & -.8796473 & -1.109216 \\ -.8796473 & 3.495621 & 0 \\ -1.109216 & 0 & 9.921136 \end{bmatrix}$$

$$\begin{bmatrix} -17.46169 & -.8789265 & 0 \\ -.8789265 & 3.495621 & 3.560333\text{E-}02 \\ 0 & 3.560333\text{E-}02 & 9.966067 \end{bmatrix}$$

$$\begin{bmatrix} -17.49849 & 0 & 1.489243\text{E-}03 \\ 0 & 3.532418 & 3.557217\text{E-}02 \\ 1.489243\text{E-}03 & 3.557217\text{E-}02 & 9.966067 \end{bmatrix}$$

$$\begin{bmatrix} -17.49849 & 8.233765\text{E-}06 & -1.48922\text{E-}03 \\ 8.233765\text{E-}06 & 3.532221 & 0 \\ -1.48922\text{E-}03 & 0 & 9.966265 \end{bmatrix}$$

$$\begin{bmatrix} -17.49849 & 8.233765\text{E-}06 & 0 \\ 8.233765\text{E-}06 & 3.532221 & -4.464509\text{E-}10 \\ 0 & -4.464508\text{E-}10 & 9.966265 \end{bmatrix}$$

The off-diagonal entries are now quite small, and we obtain the same eigenvalues from the diagonal that we did in Example 5 using the power method. ■

QR Algorithm

At the present time, an algorithm based on the *QR* factorization of an *invertible* matrix, discussed in Section 6.2, is often used by professionals to find eigenvalues of a matrix. A full treatment of the *QR* algorithm is beyond the scope of this text, but we give a brief description of the method. As with the Jacobi method, pencil-and-paper computations of eigenvalues using the *QR* algorithm are too cumbersome to include in this text. The routine QRFAC in LINTEK can be used to illustrate the features of the *QR* algorithm that we now describe.

Let A be a nonsingular matrix. The *QR* algorithm generates a sequence of matrices $A_1, A_2, A_3, A_4, \dots$, all having the same eigenvalues as A . To generate this sequence, let $A_1 = A$, and factor $A_1 = Q_1 R_1$, where Q_1 is the orthogonal matrix and R_1 is the upper-triangular matrix described in Section 6.2. Then let $A_2 = R_1 Q_1$, factor A_2 into $Q_2 R_2$, and set $A_3 = R_2 Q_2$. Continue in this fashion, factoring A_n into $Q_n R_n$ and setting $A_{n+1} = R_n Q_n$. Under fairly general

conditions, the matrices A_i will approach an almost upper-triangular matrix of the form

$$\begin{bmatrix} \times & \times & \times & \times & \times & \cdots & \times & \times \\ \times & \times & \times & \times & \times & \cdots & \times & \times \\ 0 & 0 & \times & \times & \times & \cdots & \times & \times \\ 0 & 0 & \times & \times & \times & \cdots & \times & \times \\ 0 & 0 & 0 & 0 & \times & \cdots & \times & \times \\ 0 & 0 & 0 & 0 & \times & \cdots & \times & \times \\ & & & & & \ddots & & \\ & & & & & & \ddots & \\ 0 & 0 & 0 & 0 & 0 & \cdots & \times & \times \end{bmatrix}$$

The colored entries just below the main diagonal may or may not be zero. If one of these entries is nonzero, the 2×2 submatrix having the entry in its lower left-hand corner, like the matrix shaded above, has a pair of complex conjugate numbers $a \pm bi$ as eigenvalues that are also eigenvalues of the large matrix, and of A . Entries on the diagonal that do not lie in such a 2×2 block are real eigenvalues of the matrix and of A .

The routine QRFACTOR in LINTEK can be used to illustrate this procedure. A few comments about the procedure and the program are in order.

From $A_1 = Q_1 R_1$, we have $R_1 = Q_1^{-1} A_1$. Then $A_2 = R_1 Q_1 = Q_1^{-1} A_1 Q_1$, so we see that A_2 is similar to $A_1 = A$ and therefore has the same eigenvalues as A . Continuing in this fashion, we see that each matrix A_i is similar to A . This explains why the eigenvalues don't change as the matrices of the sequence are generated.

Notice, too, that Q_i and Q_i^{-1} are orthogonal matrices, so $Q_i^{-1}x$ and yQ_i have the same magnitudes as the vectors x and y , respectively. It follows that, if E is the matrix of errors in the entries of A , then the error matrix $Q_i^{-1}EQ_i$ arising in the computation of $Q_i^{-1}AQ_i$ is of magnitude comparable to that of E . That is, the generation of the sequence of matrices A_i is *stable*. This is highly desirable in numerical computations.

Finally, it is often useful to perform a shift, adding a scalar multiple rI of the identity matrix to A_i before generating the next matrix A_{i+1} . Such a shift increases all eigenvalues by r (see Exercise 32 in Section 5.1), but we can keep track of the total change due to such shifts and adjust the eigenvalues of the final matrix found to obtain those of A . To illustrate one use of shifts, suppose that we wish to find eigenvalues of a singular matrix B . We can form an initial shift, perhaps taking $A = B + (.001)I$, to obtain an invertible matrix A to start the algorithm. For another illustration, we can find by using the routine QRFACTOR that the QR algorithm applied to the matrix

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

generates this same matrix repeatedly, even though the eigenvalues are 1 and -1 rather than complex numbers. This is an example of a matrix A for which

the sequence of matrices A_i does not approach a form described earlier. However, a shift that adds $(.9)I$ produces the matrix

$$\begin{bmatrix} .9 & 1 \\ 1 & .9 \end{bmatrix},$$

which generates a sequence that quickly converges to

$$\begin{bmatrix} 1.9 & 0 \\ 0 & -.1 \end{bmatrix}.$$

Subtracting the scalar .9 from the eigenvalues 1.9 and $-.1$ of this last matrix, we obtain the eigenvalues 1 and -1 of the original matrix.

Shifts can also be used to speed convergence, which is quite fast when the ratios of magnitudes of eigenvalues are large. The routine QRFACTOR displays the matrices A_i as they are generated, and it allows shifts. If we notice that we are going to obtain an eigenvalue whose decimal expansion starts with 17.52, then we can speed convergence greatly by adding the shift $(-17.52)I$. The resulting eigenvalue will be near zero, and the ratios of the magnitudes of other eigenvalues to its magnitude will probably be large. Using this technique with QRFACTOR, it is quite easy to find all eigenvalues, both real and complex, of most matrices of reasonable size that can be displayed conveniently.

Professional programs make many further improvements in the algorithm we have presented, in order to speed the creation of zeros in the lower part of the matrix.

SUMMARY

1. Let A be an $n \times n$ diagonalizable matrix with real eigenvalues λ_j and with a dominant eigenvalue λ_1 of algebraic multiplicity 1, so that $|\lambda_1| > |\lambda_j|$ for $j = 2, 3, \dots, n$. Start with any vector w_1 in $\mathbb{R}^n$ that is not in the subspace generated by eigenvectors corresponding to the eigenvalues λ_j for $j > 1$. Form the vectors $w_2 = (Aw_1)/d_1$, $w_3 = (Aw_2)/d_2$, $\dots$, where d_i is the maximum of the magnitudes of the components of Aw_i . The sequence of vectors

$$w_1, w_2, w_3, \dots$$

approaches an eigenvector of A corresponding to λ_1 , and the associated Rayleigh quotients $(Aw_i \cdot w_i)/(w_i \cdot w_i)$ approach λ_1 . This is the foundation for the power method, which is summarized in the box before Example 1.

2. If A is diagonalizable and invertible, and if $|\lambda_n| < |\lambda_i|$ for $i < n$ with λ_n of algebraic multiplicity 1, then the power method may be used with A^{-1} to find λ_n .

- Let A be an $n \times n$ symmetric matrix with eigenvalues λ_i such that $|\lambda_1| > |\lambda_2| \geq \dots \geq |\lambda_n|$. If $\mathbf{b}_1$ is a unit eigenvector corresponding to λ_1 , then $A - \lambda_1 \mathbf{b}_1 \mathbf{b}_1^T$ has eigenvalues $\lambda_2, \lambda_3, \dots, \lambda_n, 0$, and if $|\lambda_2| > |\lambda_3|$, then λ_2 can be found by applying the power method to $A - \lambda_1 \mathbf{b}_1 \mathbf{b}_1^T$. This deflation can be continued with $A - \lambda_1 \mathbf{b}_1 \mathbf{b}_1^T - \lambda_2 \mathbf{b}_2 \mathbf{b}_2^T$ if $|\lambda_3| > |\lambda_4|$, and so on, to find more eigenvalues.
- In the Jacobi method for diagonalizing a symmetric matrix A , one generates a sequence of symmetric matrices, starting with A . Each matrix of the sequence is obtained from the preceding one by multiplying it on the left by R^T and on the right by R , where R is an orthogonal "rotation" matrix designed to annihilate the two (symmetrically located) entries of maximum magnitude off the diagonal. The matrices in the sequence approach a diagonal matrix D having the same eigenvalues as A .
- In the QR method, one begins with an invertible matrix A and generates a sequence of matrices A_i by setting $A_1 = A$ and $A_{n+1} = R_n Q_n$, where the QR factorization of A_n is $Q_n R_n$. The matrices A_i approach almost upper-triangular matrices having the real eigenvalues of A on the diagonal and pairs of complex conjugate eigenvalues of A as eigenvalues of 2×2 blocks appearing along the diagonal. Shifts may be used to speed convergence or to find eigenvalues of a singular matrix.

EXERCISES

In Exercises 1–4, use the power method to estimate the eigenvalue of maximum magnitude and a corresponding eigenvector for the given matrix. Start with first estimate

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

and compute $\mathbf{w}_2, \mathbf{w}_3$, and $\mathbf{w}_4$. Also find the three Rayleigh quotients. Then find the exact eigenvalues, for comparison, using the characteristic equation.

- $\begin{bmatrix} 3 & -3 \\ -5 & 1 \end{bmatrix}$
- $\begin{bmatrix} 3 & -3 \\ 4 & -5 \end{bmatrix}$
- $\begin{bmatrix} -3 & 10 \\ -3 & 8 \end{bmatrix}$
- $\begin{bmatrix} -4 & 9 \\ -2 & 5 \end{bmatrix}$

In Exercises 5–8, find the spectral decomposition (8) of the given symmetric matrix.

- $\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$
- $\begin{bmatrix} 3 & 5 \\ 5 & 3 \end{bmatrix}$

$$7. \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$8. \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}$$

[HINT: Use Example 4 in Section 6.3.]

In Exercises 9–12, find the matrix obtained by deflation after the (exact) eigenvalue of maximum magnitude and a corresponding eigenvector are found.

- The matrix in Exercise 5
- The matrix in Exercise 6
- The matrix in Exercise 7
- The matrix in Exercise 8



In Exercises 13–15, use the routine **POWER** in **LINTEK**, or review Exercises M2–M5 in Section 5.1 and use **MATLAB** as in M4 and M5, to find

the eigenvalue of maximum magnitude and a corresponding eigenvector of the given matrix.

13.
$$\begin{bmatrix} 1 & -44 & -88 \\ -5 & 55 & 113 \\ 1 & -24 & -48 \end{bmatrix}$$

14.
$$\begin{bmatrix} 57 & -2 & 31 \\ -205 & 8 & -113 \\ -130 & 4 & -70 \end{bmatrix}$$

15.
$$\begin{bmatrix} 3 & -22 & -46 \\ -3 & 23 & 47 \\ 1 & -10 & -20 \end{bmatrix}$$

16. Use the software described for Exercises 13–15 with matrix inversion to find the eigenvalue of minimum magnitude and the corresponding eigenvector for the matrix in Exercise 13.

17. Repeat Exercise 16 for the matrix

$$\begin{bmatrix} -1 & 3 & 1 \\ 3 & 2 & -11 \\ 1 & -11 & 7 \end{bmatrix}$$

In Exercises 18–21, use the routine *POWER* in *LINTEK* and deflation to find all eigenvalues and corresponding eigenvectors for the given symmetric matrix. Always continue the method before deflating until as much stabilization as possible is achieved. Note the relationship between ratios of magnitudes of eigenvalues and the speed of convergence.

18.
$$\begin{bmatrix} 3 & 5 & -7 \\ 5 & 10 & 11 \\ -7 & 11 & 0 \end{bmatrix}$$

19.
$$\begin{bmatrix} 0 & -1 & 4 \\ -1 & 2 & -1 \\ 4 & -1 & 0 \end{bmatrix}$$

20.
$$\begin{bmatrix} 3 & 1 & -2 & 1 \\ 1 & 4 & 0 & -3 \\ -2 & 0 & 0 & 5 \\ 1 & -3 & 5 & 2 \end{bmatrix}$$

21.
$$\begin{bmatrix} 5 & 7 & -2 & 6 \\ 7 & 4 & 11 & 3 \\ -2 & 11 & -8 & 0 \\ 6 & 3 & 0 & 6 \end{bmatrix}$$

22. The eigenvalue option of the routine *VECTGRPH* in *LINTEK* is designed to

strengthen your geometric understanding of the power method. There is no scaling, but you have the option of starting again if the numbers get so large that the vectors are off the screen. Read the directions in the program, and then run this eigenvalue option until you can reliably achieve a score of 85% or better.

23. Consider the matrix

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

The following steps will form an orthogonal diagonalizing matrix C for A .

Step 1 Let $g = (a - c)/2$.

Step 2 Let $h = \sqrt{g^2 + b^2}$.

Step 3 Let $r = \sqrt{b^2 + (g + h)^2}$.

Step 4 Let $s = \sqrt{b^2 + (g - h)^2}$.

Step 5 Let $C = \begin{bmatrix} -b/r & -b/s \\ (g + h)/r & (g - h)/s \end{bmatrix}$.

(If $b < 0$ and a rotation matrix is desired, change the sign of a column vector in C .) Prove this algorithm by proving the following.

- a. The eigenvalues of A are

$$\lambda = \frac{a + c \pm \sqrt{(a - c)^2 + 4b^2}}{2}$$

[HINT: Use the quadratic formula.]

- b. The first row vector of $A - \lambda I$ is $(g \pm \sqrt{g^2 + b^2}, b)$.
c. The columns of the orthogonal matrix C in step 5 can be found using part b.
d. The parenthetical statement following step 5 is valid.

24. Use the algorithm in Exercise 23 to find an orthogonal diagonalizing matrix with determinant 1 for each of the following.

a.
$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

b.
$$\begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix}$$

In Exercises 25–28, use the routine *JACOBI* in *LINTEK* to find the eigenvalues of the matrix by Jacobi's method.

25. The matrix in Exercise 18
 26. The matrix in Exercise 19
 27. The matrix in Exercise 20
 28. The matrix in Exercise 21

In Exercises 29–32, use the routine *QRFACTOR* in *LINTEK* to find all eigenvalues, both real and complex, of the given matrix. Be sure to make use of shifts to speed convergence whenever you can see approximately what an eigenvalue will be.

$$29. \begin{bmatrix} -1 & 5 & 7 \\ 3 & -6 & 8 \\ 9 & 0 & -11 \end{bmatrix}$$

$$30. \begin{bmatrix} -3 & 6 & -7 & 2 \\ 13 & -18 & 4 & 6 \\ 21 & 32 & -16 & 9 \\ -4 & 8 & 7 & 11 \end{bmatrix}$$

$$31. \begin{bmatrix} -12 & 3 & 15 & 2 & -21 \\ 47 & -34 & 87 & 24 & 7 \\ 35 & 72 & 33 & -57 & 82 \\ -145 & 67 & 32 & 10 & 46 \\ -9 & 22 & 21 & -45 & 8 \end{bmatrix}$$

$$32. \begin{bmatrix} -3 & 6 & 4 & -2 & 16 \\ 21 & -33 & 5 & 8 & -12 \\ 15 & -21 & 13 & 4 & 20 \\ -18 & 12 & 4 & 8 & 3 \\ -22 & 31 & 14 & 9 & 10 \end{bmatrix}$$

COMPLEX SCALARS

Much of the linear algebra we have discussed is equally valid in applications using complex numbers as scalars. In fact, the use of complex scalars actually extends and clarifies many of the results we have developed. For example, when we are dealing with complex scalars, every $n \times n$ matrix has n eigenvalues, counted with their algebraic multiplicities.

In Section 9.1, we review the algebra of complex numbers. Section 9.2 summarizes properties of the complex vector space $\mathbb{C}^n$ and of matrices with complex entries. Diagonalization of these matrices and Schur's unitary triangularization theorem are discussed in Section 9.3. Section 9.4 is devoted to Jordan canonical forms.

9.1

ALGEBRA OF COMPLEX NUMBERS

The Number i

Numbers exist only in our minds. There is no physical entity that *is* the number 1. If there were, 1 would be in a place of high honor in some great museum of science, and past it would file a steady stream of mathematicians, gazing at 1 in wonder and awe. As mathematics developed, new numbers were *invented* (some prefer to say *discovered*) to fulfill algebraic needs that arose. If we start just with the positive integers, which are the most *natural numbers*, inventing *zero* and *negative integers* enables us not merely to add any two integers, but also to subtract any two integers. Inventing *rational numbers* (fractions) enables us to divide an integer by any nonzero integer. It can be shown that no rational number squared is equal to 2, so *irrational numbers* have to be invented to solve the equation $x^2 = 2$, and our familiar decimal notation comes into use. This decimal notation provides us with other numbers, such as π and e , that are of practical use. The numbers (positive, negative, and zero) that we customarily write using decimal notation have unfortunately become known as the *real numbers*; they are no more real than any other numbers we may invent, because all numbers exist only in our minds.

The real numbers are still inadequate to provide solutions of even certain polynomial equations. The simple equation $x^2 + 1 = 0$ has no real number as a solution. In terms of our needs in this text, the matrix

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

has no eigenvalues. Mathematicians have had to invent a solution i of the equation $x^2 + 1 = 0$ to fill this need. Of course, once a new number is invented, it must take its place in the algebra of numbers; that is, we would like to multiply it and add it to all the numbers we already have. Thus, we also allow numbers bi and finally $a + bi$ for all of our old real numbers a and b . As we will subsequently show, it then becomes possible to add, subtract, multiply, and divide any of these new *complex numbers*, except that division by zero is still not possible. It is unfortunate that i has become known as an *imaginary number*. The purpose of this introduction is to point out that i is no less real and no more imaginary than any other number.

With the invention of i and the consequent enlargement of our number system to the set $\mathbb{C}$ of all the complex numbers $a + bi$, a marvelous thing happens. Not only does the equation $x^2 + a = 0$ have a solution for all real numbers a , but *every* polynomial equation has a solution in the complex numbers. We state without proof the famous *Fundamental Theorem of Algebra*, which shows this.

Fundamental Theorem of Algebra

Every polynomial equation with coefficients in $\mathbb{C}$ has n solutions in $\mathbb{C}$, where n is the degree of the polynomial and the solutions are counted with their algebraic multiplicity.

Review of Complex Arithmetic

A **complex number** z is any expression $z = a + bi$, where a and b are real numbers and $i = \sqrt{-1}$. The scalar a is the **real part** of z , and b is the **imaginary part** of z . It is useful to represent the complex number $z = a + bi$ as the vector $[a, b]$ in the x, y -plane $\mathbb{R}^2$, as shown in Figure 9.1. The x -axis is the **real axis**, and the y -axis is the **imaginary axis**. We let $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$ be the set of all complex numbers.

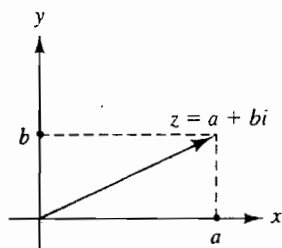


FIGURE 9.1
The complex number z .

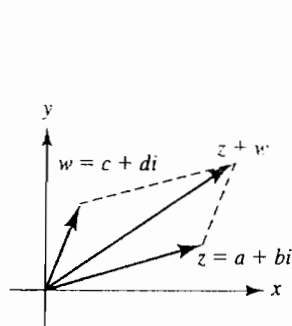


FIGURE 9.2
Representation of $z + w$.

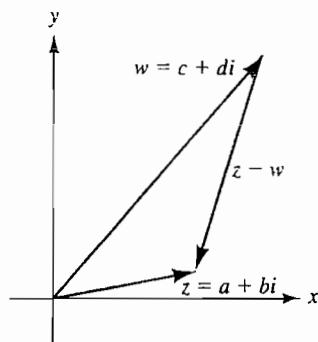


FIGURE 9.3
Representation of $z - w$.

Complex numbers are added, subtracted, and multiplied by real scalars in the natural way:

$$\underbrace{(a + bi)}_z \pm \underbrace{(c + di)}_w = (a \pm c) + (b \pm d)i,$$

$$r(a + bi) = ra + (rb)i.$$

These operations have the same geometric representations as those for vectors in $\mathbb{R}^2$ and are illustrated in Figures 9.2, 9.3, and 9.4.

It is clear that the set $\mathbb{C}$ of complex numbers is a *real* vector space of dimension 2, isomorphic to $\mathbb{R}^2$.

The **modulus** (or **magnitude**) of the complex number $z = a + bi$ is $|z| = \sqrt{a^2 + b^2}$, which is the length of the vector in Figure 9.1. Notice that $|z| = 0$ if and only if $z = 0$ —that is, if and only if $a = b = 0$.

Multiplication of two complex numbers is defined in the way it has to be to make the distributive laws $z_1(z_2 + z_3) = z_1z_2 + z_1z_3$ and $(z_1 + z_2)z_3 = z_1z_3 + z_2z_3$ valid. Namely, remembering that $i^2 = -1$, we have

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

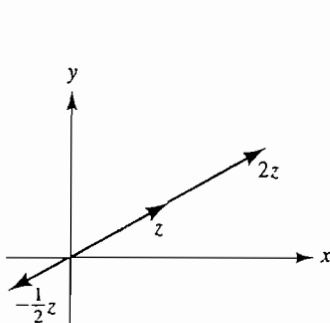


FIGURE 9.4
Representation of rz .

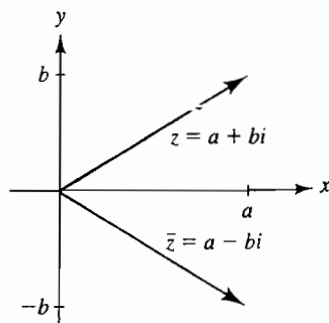


FIGURE 9.5
The conjugate of z .

Multiplication of complex numbers is commutative and associative. To divide one complex number by a nonzero complex number, we make use of the notion of a *complex conjugate*. The **conjugate** of $z = a + bi$ is $\bar{z} = a - bi$. Figure 9.5 illustrates the geometric relationship between $\bar{z}$ and z . Computing $z\bar{z} = (a + bi)(a - bi)$, we find that

$$z\bar{z} = a^2 + b^2 = |z|^2. \quad (1)$$

If $z \neq 0$, then Eq. (1) can be written as $z(\bar{z}/|z|^2) = 1$, so $1/z = \bar{z}/|z|^2$. More generally, if $z \neq 0$, then w/z is computed as

$$\frac{w}{z} = \frac{w\bar{z}}{z\bar{z}} = \frac{w\bar{z}}{|z|^2} = \frac{1}{|z|^2}(w\bar{z}). \quad (2)$$

We will see geometric representations of multiplication and division in a moment.

EXAMPLE 1 Find $(3 + 4i)^{-1} = 1/(3 + 4i)$.

SOLUTION Using Eq. (2), we have

$$\frac{1}{3 + 4i} = \left(\frac{1}{3 + 4i} \right) \left(\frac{3 - 4i}{3 - 4i} \right) = \frac{3 - 4i}{25} = \frac{1}{25}(3 - 4i) = \frac{3}{25} - \frac{4}{25}i. \quad \blacksquare$$

EXAMPLE 2 Compute $(2 + 3i)/(1 + 2i)$.

SOLUTION Using the technique of Eq. (2), we obtain

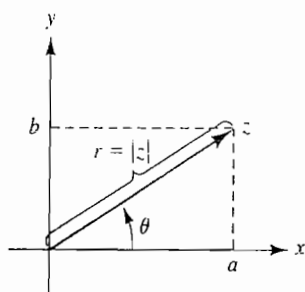
$$\frac{2 + 3i}{1 + 2i} = \left(\frac{2 + 3i}{1 + 2i} \right) \left(\frac{1 - 2i}{1 - 2i} \right) = \frac{1}{5}(2 + 3i)(1 - 2i) = \frac{8}{5} - \frac{1}{5}i. \quad \blacksquare$$

All of the familiar rules of arithmetic apply to the algebra of complex numbers. However, there is no notion of *order* for complex numbers; the notion $z_1 < z_2$ is defined only if z_1 and z_2 are real. The conjugation operation turns out to be very important; we summarize its properties in a theorem.

THEOREM 9.1 Properties of Conjugation in $\mathbb{C}$

Let $z = a + bi$ and $w = c + di$ be complex numbers. Then

1. $\overline{z + w} = \bar{z} + \bar{w}$,
2. $\overline{z - w} = \bar{z} - \bar{w}$,
3. $\overline{z\bar{w}} = \bar{z}w$,
4. $\overline{z/w} = \bar{z}/\bar{w}$,
5. $\overline{\bar{z}} = z$.

FIGURE 9.6
Polar form of z .

The proofs of the properties in Theorem 9.1 are very easy. We prove property 3 as an example and leave the rest as exercises. (See Exercises 12 and 13.)

EXAMPLE 3 Prove property 3 of Theorem 9.1.

SOLUTION Let $z = a + bi$ and $w = c + di$. Then

$$\begin{aligned}\bar{z}\bar{w} &= (\overline{a + bi})(\overline{c + di}) = (a - bi)(c - di) \\ &= (ac - bd) - (ad + bc)i = \overline{zw}.\end{aligned}$$

Polar Form of Complex Numbers

Let us return to the vector representation of $z = a + bi$ in $\mathbb{R}^2$. Figure 9.6 indicates that, for $z \neq 0$, if θ is an angle from the positive x -axis to the vector representation of z , and if $r = |z|$, then $a = r \cos \theta$ and $b = r \sin \theta$. Thus,

$$z = r(\cos \theta + i \sin \theta). \quad \text{A polar form of } z \quad (3)$$

HISTORICAL NOTE COMPLEX NUMBERS MADE THEIR INITIAL APPEARANCE ON the mathematical scene in *The Great Art, or On the Rules of Algebra* (1545) by the 16th century Italian mathematician and physician, Gerolamo Cardano (1501–1576). It was in this work that Cardano presented an algebraic method of solution of cubic and quartic equations. But it was the quadratic problem of dividing 10 into two parts such that their product is 40 to which Cardano found the solution $5 + \sqrt{-15}$ and $5 - \sqrt{-15}$ by standard techniques. Cardano was not entirely happy with this answer, as he wrote, “So progresses arithmetic subtlety the end of which, as is said, is as refined as it is useless.”

Twenty-seven years later the engineer Raphael Bombelli (1526–1572) published an algebra text in which he dealt systematically with complex numbers. Bombelli wanted to clarify Cardano's cubic formula, which under certain circumstances could express a correct real solution of a cubic equation as the sum of two expressions each involving the square root of a negative number. Thus he developed our modern rules for operating with expressions of the form $a + b\sqrt{-1}$, including methods for determining cube roots of such numbers. Thus, for example, he showed that $\sqrt[3]{2 + \sqrt{-121}} = 2 + \sqrt{-1}$ and therefore that the solution to the cubic equation $x^3 = 15x + 4$, which Cardano's formula gave as $x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}}$, could be written simply as $x = (2 + \sqrt{-1}) + (2 - \sqrt{-1})$, or as $x = 4$, the obvious answer.

The angle θ is an **argument** of z . Of particular importance is the restricted value

$$-\pi < \theta \leq \pi \quad \text{The principal argument of } z$$

denoted by $\text{Arg}(z)$. We usually use this **principal argument** of z and refer to the corresponding form (3) as **the polar form** of z .

EXAMPLE 4 Find the principal argument and the polar form of the complex number $z = -\sqrt{3} - i$.

SOLUTION Because $|z| = \sqrt{(-\sqrt{3})^2 + (-1)^2} = \sqrt{4} = 2$, we have $\cos \theta = -\sqrt{3}/2$ and $\sin \theta = -1/2$, as indicated in Figure 9.7. The principal argument of z is the angle θ between $-\pi$ and π satisfying these two conditions—that is, $\theta = -5\pi/6$. The required polar form is $z = 2(\cos(-5\pi/6) + i \sin(-5\pi/6))$. ■

If we multiply two complex numbers in their polar form, we quickly discover the geometric representation of multiplication. Suppose that we let $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$. Then

$$\begin{aligned} z_1 z_2 &= r_1 r_2 ((\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)) \\ &= r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)). \end{aligned}$$

The last equation arises from the familiar trigonometric identities for the cosine and sine of a sum. This computation shows that, when two complex numbers are multiplied, the moduli are multiplied and the arguments are added. Notice, however, that the principal argument of a product may not be the sum of the principal arguments of the factors; the sum of the principal arguments may have to be adjusted to lie between $-\pi$ and π . (See Figure 9.8.)

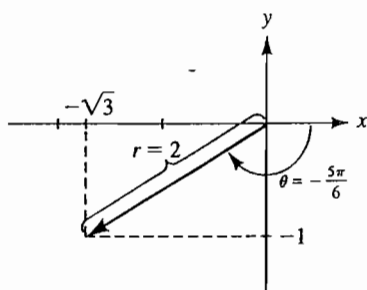


FIGURE 9.7
The polar form of $z = -\sqrt{3} - i$.

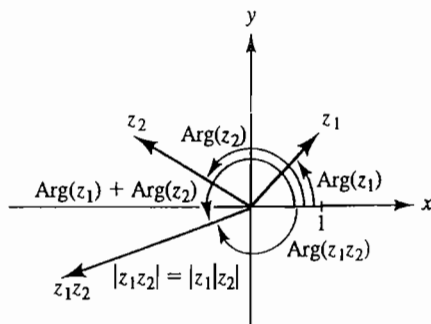


FIGURE 9.8
Representation of $z_1 z_2$.

Geometric Representation of $z_1 z_2$

1. $|z_1 z_2| = |z_1| |z_2|$.
 2. $\text{Arg}(z_1) + \text{Arg}(z_2)$ is an argument of $z_1 z_2$.
- (4)

Because z_1/z_2 is the complex number w such that $z_2 w = z_1$, we see that, in division, one divides the moduli and subtracts the arguments. (See Figure 9.9.)

Geometric Representation of z_1/z_2

1. $|z_1/z_2| = |z_1|/|z_2|$.
 2. $\text{Arg}(z_1) - \text{Arg}(z_2)$ is an argument of z_1/z_2 .
- (5)

EXAMPLE 5 Illustrate relations (4) and (5) for the complex numbers $z_1 = \sqrt{3} + i$ and $z_2 = 1 + i$.

SOLUTION For z_1 , we have $r_1 = |z_1| = \sqrt{4} = 2$, $\cos \theta_1 = \sqrt{3}/2$, and $\sin \theta_1 = 1/2$, so

$$z_1 = 2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}).$$

For z_2 , we have $r_2 = |z_2| = \sqrt{2}$ and $\cos \theta_2 = \sin \theta_2 = 1/\sqrt{2}$, so

$$z_2 = \sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}).$$

Thus,

$$\begin{aligned} z_1 z_2 &= 2\sqrt{2}(\cos(\frac{\pi}{6} + \frac{\pi}{4}) + i \sin(\frac{\pi}{6} + \frac{\pi}{4})) \\ &= 2\sqrt{2}(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12}), \end{aligned}$$

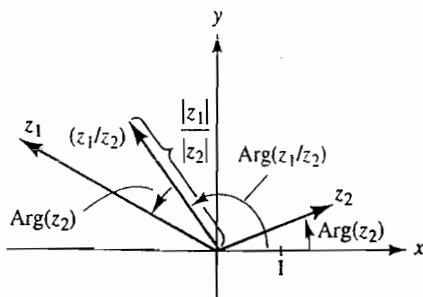


FIGURE 9.9
Representation of z_1/z_2 .

and

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{2}{\sqrt{2}} \left(\cos\left(\frac{\pi}{6} - \frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{6} - \frac{\pi}{4}\right) \right) \\ &= \frac{2}{\sqrt{2}} \left(\cos\left(-\frac{\pi}{12}\right) + i \sin\left(-\frac{\pi}{12}\right) \right).\end{aligned}$$

Relations (4) can be used repeatedly with $z_1 = z_2 = z = r(\cos \theta + i \sin \theta)$ to compute z^n for a positive integer n . We obtain the formula

$$z^n = r^n (\cos n\theta + i \sin n\theta). \quad (6)$$

We can find n th roots of z by solving the equation $w^n = z$. Writing $z = r(\cos \theta + i \sin \theta)$ and $w = s(\cos \phi + i \sin \phi)$ and using Eq. (6), the equation $z = w^n$ becomes

$$r(\cos \theta + i \sin \theta) = s^n (\cos n\phi + i \sin n\phi).$$

Therefore, $r = s^n$ and $n\phi = \theta \pm 2k\pi$ for $k = 0, 1, 2, \dots$, so

$$w = s(\cos \phi + i \sin \phi),$$

where

$$s = r^{1/n} \quad \text{and} \quad \phi = \frac{\theta}{n} \pm \frac{2k\pi}{n}.$$

Exercise 27 indicates that these values for ϕ represent precisely n distinct complex numbers, as indicated in the following box.

n th roots of $z = r(\cos \theta + i \sin \theta)$

The n th roots of z are

$$r^{1/n} \left(\cos \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) + i \sin \left(\frac{\theta}{n} + \frac{2k\pi}{n} \right) \right) \quad (7)$$

for $k = 0, 1, 2, \dots, n-1$.

This illustrates the Fundamental Theorem of Algebra for the equation $w^n = z$ of degree n that has n solutions in $\mathbb{C}$.

EXAMPLE 6 Find the fourth roots of 16, and draw their vector representations in $\mathbb{R}^2$.

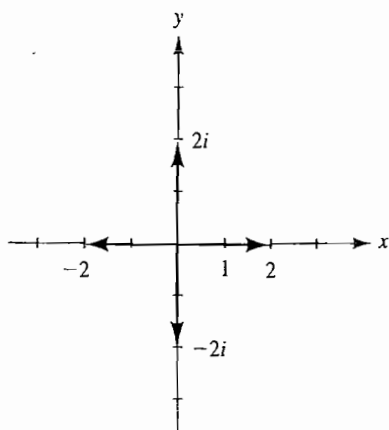


FIGURE 9.10
The fourth roots of 16.

SOLUTION The polar form of $z = 16$ is $z = 16(\cos 0 + i \sin 0)$, where $\text{Arg}(z) = 0$. Applying formula (7), with $n = 4$, we find the following fourth roots of 16 (see Figure 9.10):

k	$16^{1/4} \left(\cos \frac{2k\pi}{4} + i \sin \frac{2k\pi}{4} \right)$
0	$2(\cos 0 + i \sin 0) = 2$
1	$2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) = 2i$
2	$2(\cos \pi + i \sin \pi) = -2$
3	$2 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) = -2i$

SUMMARY

1. A complex number is a number of the form $z = a + bi$, where a and b are real numbers and $i = \sqrt{-1}$.
2. The modulus of the complex number $z = a + bi$ is

$$|z| = \sqrt{a^2 + b^2}.$$
3. The complex number $z = a + bi$ has the polar form $z = |z|(\cos \theta + i \sin \theta)$, and $\theta = \text{Arg}(z)$ is the principal argument of z if $-\pi < \theta \leq \pi$.

4. Arithmetic computations in the set $\mathbb{C}$ of complex numbers, including division and extraction of roots, can be accomplished as illustrated in this section and can be represented geometrically as shown in the figures of this section.

EXERCISES

- Find the sum $z + w$ and the product zw if
 - $z = 1 + 2i$, $w = 3 - i$,
 - $z = 3 + i$, $w = i$.
- Find the sum $z + w$ and the product zw if
 - $z = 2 + 3i$, $w = 5 - i$,
 - $z = 1 + 2i$, $w = 2 - i$.
- Find $|z|$ and $\bar{z}$, and verify that $z\bar{z} = |z|^2$, if
 - $z = 3 + 2i$,
 - $z = 4 - i$.
- Find $|z|$ and $\bar{z}$, and verify that $z\bar{z} = |z|^2$, if
 - $z = 2 + i$,
 - $z = 3 - 4i$.
- Show that z is a real number if and only if $z = \bar{z}$.
- Express z^{-1} in the form $a + bi$, for a and b real numbers, if
 - $z = -1 + i$,
 - $z = 3 + 4i$.
- Express z/w in the form $a + bi$, for a and b real numbers, if
 - $z = 1 + 2i$, $w = 1 + i$,
 - $z = 3 + i$, $w = 3 + 4i$.
- Find the modulus and principal argument for
 - $\sqrt{3} - i$,
 - $-\sqrt{3} - i$.
- Find the modulus and principal argument for
 - $-2 + 2i$,
 - $-2 - 2i$.
- Express $(\sqrt{3} + i)^6$ in the form $a + bi$ for a and b real numbers. [HINT: Write the given number in polar form.]
- Express $(1 + i)^8$ in the form $a + bi$ for a and b real numbers. [HINT: Write the given number in polar form.]
- Prove properties 1, 2, and 5 of Theorem 9.1.
- Prove property 4 of Theorem 9.1.
- Illustrate Eqs. (5) in the text for $z_1 = \sqrt{3} + i$ and $z_2 = -1 + \sqrt{3}i$.
- Illustrate Eqs. (5) in the text for $z_1 = 2 + 2i$ and $z_2 = 1 + \sqrt{3}i$.
- If $z^8 = 16$, find $|z|$.
- Mark each of the following True or False.
 - The existence of complex numbers is more doubtful than the existence of real numbers.
 - Pencil-and-paper computations with complex numbers are more cumbersome than with real numbers.
 - The square of every complex number is a positive real number.
 - Every complex number has two distinct square roots in $\mathbb{C}$.
 - Every nonzero complex number has two distinct square roots in $\mathbb{C}$.
 - The Fundamental Theorem of Algebra asserts that the algebraic operations of addition, subtraction, multiplication, and division are possible with any two complex numbers, as long as we do not divide by zero.
 - The product of two complex numbers cannot be a real number unless both numbers are themselves real or unless both are of the form bi , where b is a real number.
 - If $(a + bi)^3 = 8$, then $a^2 + b^2 = 4$.
 - If $\text{Arg}(z) = 3\pi/4$ and $\text{Arg}(w) = -\pi/2$, then $\text{Arg}(z/w) = 5\pi/4$.
 - If $z + \bar{z} = 2z$, then z is a real number.
- Find the three cube roots of 8.
- Find the four fourth roots of -16 .
- Find the three cube roots of -27 .

21. Find the four fourth roots of 1.
22. Find the six sixth roots of 1.
23. Find the eight eighth roots of 256.
24. A **primitive n th root of unity** is a complex number z such that $z^n = 1$ but $z^m \neq 1$ for $m < n$.
- Give a formula for one primitive n th root of unity.
 - Find the primitive fourth roots of unity.
 - How many primitive eighth roots of unity are there? [HINT: Argue geometrically in terms of polar forms.]
25. Let $z, w \in \mathbb{C}$. Show that $|z + w| \leq |z| + |w|$. [HINT: Remember that $\mathbb{C}$ is a real vector space of dimension 2, naturally isomorphic to $\mathbb{R}^2$.]
26. Show that the n th roots of $z \in \mathbb{C}$ can be represented geometrically as n equally spaced points on the circle $x^2 + y^2 = |z|^2$.
27. Show that the infinite list of values $\phi = \frac{\theta}{n} \pm \frac{2k\pi}{n}$ for $k = 0, 1, 2, \dots$ yields just n distinct complex numbers $w = s(\cos \phi + i \sin \phi)$ of modulus s .
28. In calculus it is shown that
- $$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$
- $$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots$$
- $$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$$
- Proceeding formally, show that $e^{i\theta} = \cos \theta + i \sin \theta$. (This is Euler's formula.)
 - Show that every complex number z may be written in the form $z = re^{i\theta}$, where $r = |z|$ and $\theta = \text{Arg}(z)$.
 - Assuming that the usual laws of exponents hold for exponents that are complex numbers, use part b to derive again Eqs. (4) and (5) describing multiplication and division of complex numbers in polar form.

9.2

MATRICES AND VECTOR SPACES WITH COMPLEX SCALARS

Complex Matrices and Linear Systems

Both the real number system $\mathbb{R}$ and the complex number system $\mathbb{C}$ are algebraic structures known as *fields*. In a field, we can add any two elements and multiply any two elements to produce an element of the field. Addition and multiplication are commutative and associative operations, and multiplication is distributive over addition. The field contains an additive identity 0 and a multiplicative identity 1. Every element c in the field has an additive inverse $-c$ in the field—that is, an element that, when added to c , produces the additive identity. Similarly, every nonzero element d in the field has a multiplicative inverse $1/d$ in the field.

The part of our work in Chapters 1, 2, and 3 that rests only on the field axioms of $\mathbb{R}$ is equally valid if we allow complex scalars. In particular, we can work with *complex matrices*—that is, with matrices having complex entries: adding matrices of the same size, multiplying matrices of appropriate sizes, and multiplying a matrix by a complex scalar. We can solve linear systems by using the same Gauss or Gauss–Jordan methods that we used in Chapter 1. All of our work in Chapter 1, Sections 3 through 6, makes perfectly good sense when applied to complex scalars. Pencil-and-paper computations are more tedious, however.

EXAMPLE 1 Solve the linear system

$$\begin{aligned} z_1 - z_2 + (1+i)z_3 &= i \\ iz_1 - 2iz_2 + iz_3 &= 2 - i \\ iz_2 - (1+i)z_3 &= 1 + 2i. \end{aligned}$$

SOLUTION We use the Gauss-Jordan method as follows:

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & -1 & 1+i & i \\ i & -2i & i & 2-i \\ 0 & i & -1-i & 1+2i \end{array} \right] &\sim \left[\begin{array}{ccc|c} 1 & -1 & 1+i & i \\ 0 & -i & 1 & 3-i \\ 0 & i & -1-i & 1+2i \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & 0 & 1+2i & 1+4i \\ 0 & 1 & i & 1+3i \\ 0 & 0 & -i & 4+i \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 10+2i \\ 0 & 1 & 0 & 5+4i \\ 0 & 0 & 1 & -1+4i \end{array} \right]. \end{aligned}$$

Thus we obtain the solution $z_1 = 10 + 2i$, $z_2 = 5 + 4i$, $z_3 = -1 + 4i$. ■

EXAMPLE 2 Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 2i & 1+i \\ 1 & 3i & i \\ 0 & 1+i & -1 \end{bmatrix}.$$

SOLUTION We proceed precisely as in Chapter 1:

$$\begin{aligned} \left[\begin{array}{ccc|ccc} 1 & 2i & 1+i & 1 & 0 & 0 \\ 1 & 3i & i & 0 & 1 & 0 \\ 0 & 1+i & -1 & 0 & 0 & 1 \end{array} \right] &\sim \left[\begin{array}{ccc|ccc} 1 & 2i & 1+i & 1 & 0 & 0 \\ 0 & i & -1 & -1 & 1 & 0 \\ 0 & 1+i & -1 & 0 & 0 & 1 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 3+i & 3 & -2 & 0 \\ 0 & 1 & i & i & -i & 0 \\ 0 & 0 & -i & 1-i & -1+i & 1 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1-4i & 4i & 1-3i \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1+i & -1-i & i \end{array} \right]. \end{aligned}$$

Thus,

$$A^{-1} = \begin{bmatrix} 1-4i & 4i & 1-3i \\ 1 & -1 & 1 \\ 1+i & -1-i & i \end{bmatrix}. \quad \blacksquare$$

Complex Vector Spaces

The definition of a *complex vector space* is identical with that of a real vector space, except that the field of scalars used is $\mathbb{C}$ rather than $\mathbb{R}$. The set $\mathbb{C}^n$ of all n -tuples having entries in $\mathbb{C}$ is an example of a complex vector space. Another example is the set of all $m \times n$ matrices with complex entries. The vector space

$\mathbb{C}^n$ has the same standard basis $\{e_1, e_2, \dots, e_n\}$ as $\mathbb{R}^n$, but of course now the field of scalars is $\mathbb{C}$. Thus, $\mathbb{C}^n$ is an n -dimensional *complex* vector space—that is, if we use *complex scalars*. In particular, $\mathbb{C} = \mathbb{C}^1$ is a one-dimensional complex vector space, even though we can regard it geometrically as a plane. That plane is a two-dimensional *real* vector space, but is a one-dimensional *complex* vector space. In general, $\mathbb{C}^n$ can be regarded geometrically in a natural way as a $2n$ -dimensional real vector space, as we ask you to explain in Exercise 1.

All of our work in Chapters 1, 2, and 3 regarding subspaces, generating sets, independence, and bases carries over to complex vector spaces, and the proofs are identical.

EXAMPLE 3 Determine whether the set

$$S = \{[1, 2i, 1 + i], [1, 3i, i], [0, 1 + i, -1]\}$$

is independent and is a basis for $\mathbb{C}^3$.

SOLUTION The vectors given in S are the row vectors in matrix A of Example 2. Row reduction of the matrix in that example shows that the matrix has rank 3, so the vectors in S are independent and hence form a basis for the three-dimensional space $\mathbb{C}^3$. ■

EXAMPLE 4 Find the coordinate vector $\mathbf{v}_B$ in $\mathbb{C}^3$ of the vector $\mathbf{v} = [i, 2 - i, 1 + 2i]$ relative to the ordered basis

$$B = \{[1, i, 0], [-1, -2i, i], [1 + i, i, -1 - i]\}.$$

SOLUTION To find the coordinate vector of $\mathbf{v}$ relative to B , we reduce the augmented matrix having the vectors in B as column vectors and having the vector $\mathbf{v}$ as the column to the right of the partition. This is precisely the augmented matrix that we reduced in Example 1, so we see that the coordinate vector $\mathbf{v}_B$, written as usual as a column vector, is

$$\mathbf{v}_B = \begin{bmatrix} 10 + 2i \\ 5 + 4i \\ -1 + 4i \end{bmatrix}. \quad \blacksquare$$

Euclidean Inner Product in $\mathbb{C}^n$

We now come to an essential difference in the structures of $\mathbb{C}$ and of $\mathbb{R}$. We have a natural idea of order for the elements of $\mathbb{R}$. We know what it means to say $x_1 < x_2$, and we have often used the fact that $x^2 \geq 0$ for all $x \in \mathbb{R}$. There is no idea of order in $\mathbb{C}$, extending the ordering of $\mathbb{R}$, for $i^2 = -1$ in $\mathbb{C}$. The nonzero numbers in $\mathbb{C}$ cannot be classified as either positive or negative on the basis of whether or not they are squares, because *all* numbers in $\mathbb{C}$ are squares. This is a very important difference between $\mathbb{R}$ and $\mathbb{C}$.

Let us see what problems this causes as we try to extend some more of the ideas in Chapter 3. We multiply matrices with complex entries by taking dot products of row vectors of the first with column vectors of the second, just as